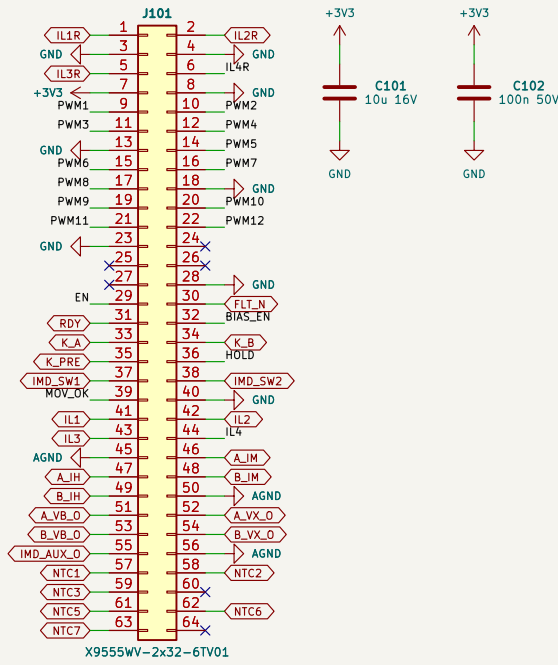


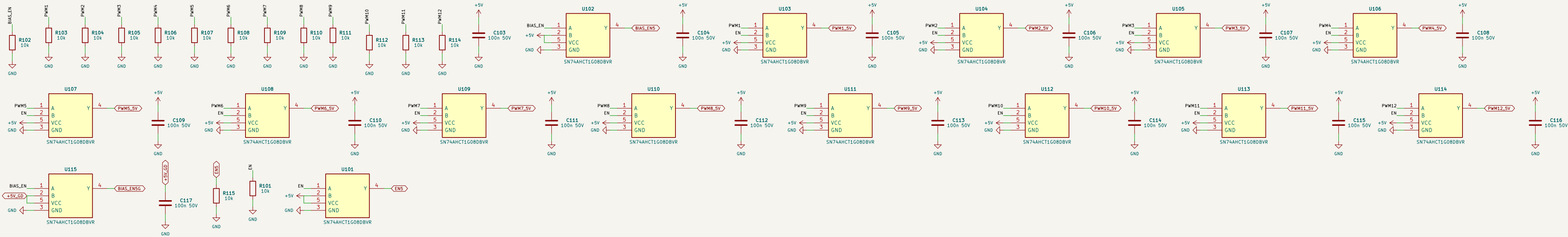
1	2	3	4	5	6	7	8
PV–PWR power board Rev A2 2026–10–05 Schematics authored by Chinmoy Bhuyan PV–P75: 3 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)							
01 PC connector, command buffers, status interfaces.PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points File: sheets/01_pc.kicad_sch 02 Phase 1 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/02_ph1_power.kicad_sch 03 Phase 1 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/03_ph1_gda.kicad_sch 04 Phase 1 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/04_ph1_gdb.kicad_sch							
05 Phase 2 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/05_ph2_power.kicad_sch 06 Phase 2 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/06_ph2_gda.kicad_sch 07 Phase 2 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/07_ph2_gdb.kicad_sch 08 Phase 3 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/08_ph3_power.kicad_sch							
09 Phase 3 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/09_ph3_gda.kicad_sch 10 Phase 3 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/10_ph3_gdb.kicad_sch 11 Port A power path (pv) Varistor Y + monitor File: sheets/11_leanA_power.kicad_sch 12 Port A sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/12_leanA_ctrl.kicad_sch							
13 Port A coil drives Contactor: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/13_leanA_drive.kicad_sch 13 Port B power path (battery) Varistor Y + monitor, aR fuses File: sheets/13_leanB_power.kicad_sch 14 Port B sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/14_leanB_ctrl.kicad_sch 15 Port B coil drives Contactor + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/15_leanB_drive.kicad_sch							
15 Terminals, EMI filter, insulation test CAUTION! no one should ever open the enclosure or touch any terminals on each build (output 24 V / PV–PWR design 6V–60VDC, 400W/800W) rev 3 from 5624.6 mW, is 8x, GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board (MD crossing), SELV (AUX04T1 from force motor (AUX–HSELV), CC output 0V). File: sheets/15_terminals_imd.kicad_sch 16 Live 5 V / 3.3 V, SELV connector Live 23.62–24.68 V (FB); SELV 20.3–29.8 V over the load table, gates off 20.3–28.2 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low File: sheets/16_supplies.kicad_sch 17 AUX75 input taps, OR, film One tap per port (+ pole, terminal side), return = BUS–; 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 350V (AUX04T1 from force motor (AUX–HSELV), CC output 0V). File: sheets/17_aux75_input.kicad_sch 18 AUX75 start–up, brown–in, OV, timer As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 35–80 ms File: sheets/18_aux75_startup.kicad_sch							
19 AUX75 PWM, SiC, clamp, AUX–T1 UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nsely 48:8:9; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding File: sheets/19_aux75_pwm_t1.kicad_sch 20 AUX75 live 24 V, SELV 24 V, status Live 23.62–24.68 V (FB); SELV 20.3–29.8 V over the load table, gates off 20.3–28.2 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low File: sheets/20_aux75_status.kicad_sch							
Not bench–validated. Values read from sim/out/*.json; checks: design_check() Sheet 1 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic Sheet: / File: PV–PWR.kicad_sch Title: PV–PWR power board – Cover / Hierarchy Size: A3Date: 2026–10–05Rev: A2 KiCad E.D.A. 10.0.6Id: 1/23							
5	6	7	8				

PC CONNECTOR (INTERFACES.PC), LIVE



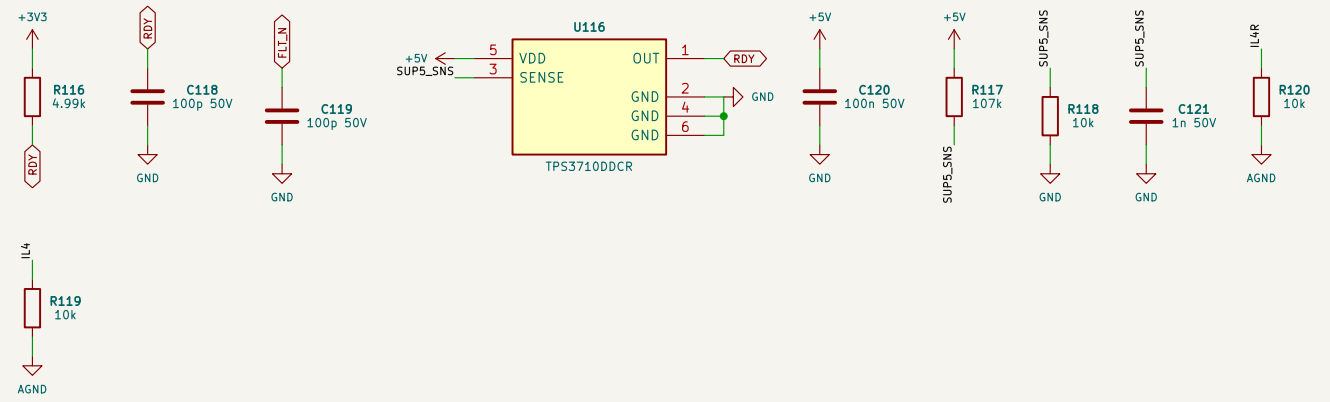
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2; the former +24V / +5V pins carry the sensor references IL1R-IL4R). Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).
PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 and IL4R pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



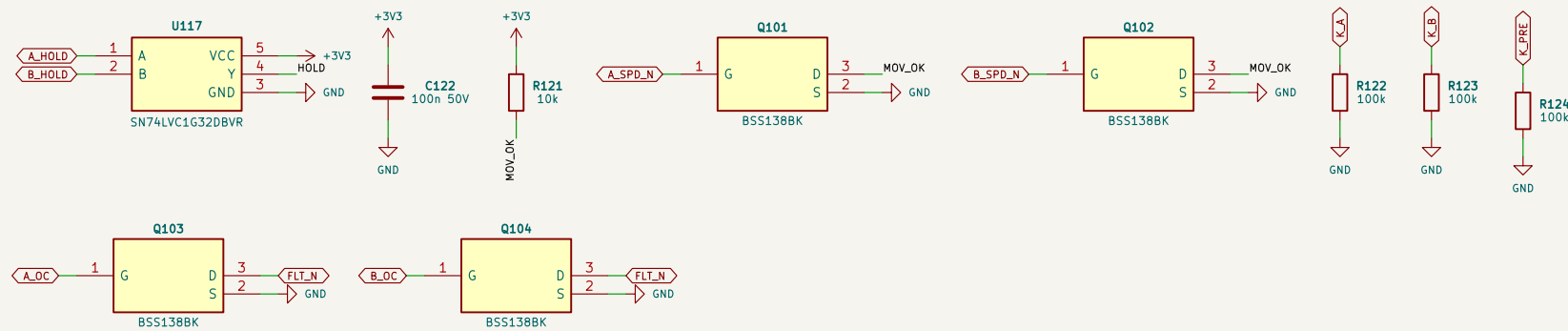
NSI6651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



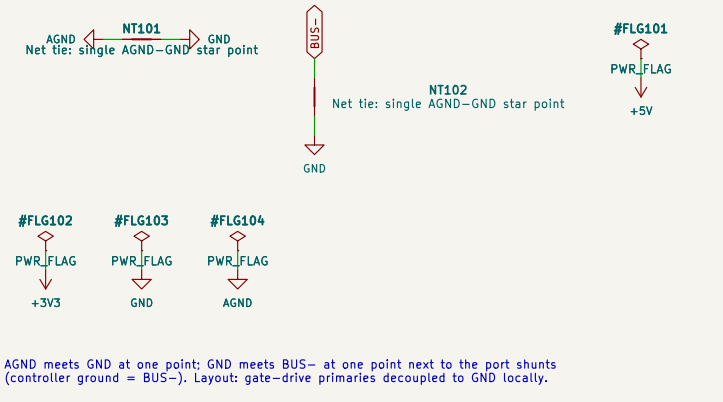
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
+5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHC2 VCC <= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



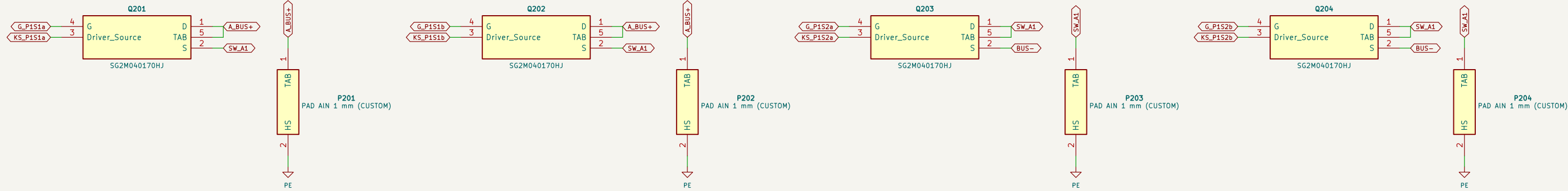
HOLD = A_HOLD OR B_HOLD ([Lport] >= 0.97-1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



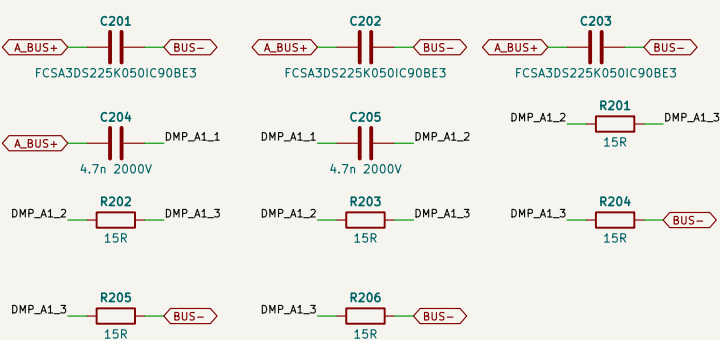
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-); Layout: gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



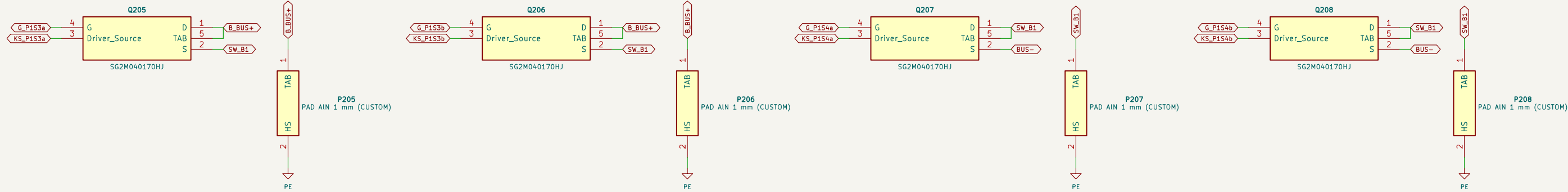
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



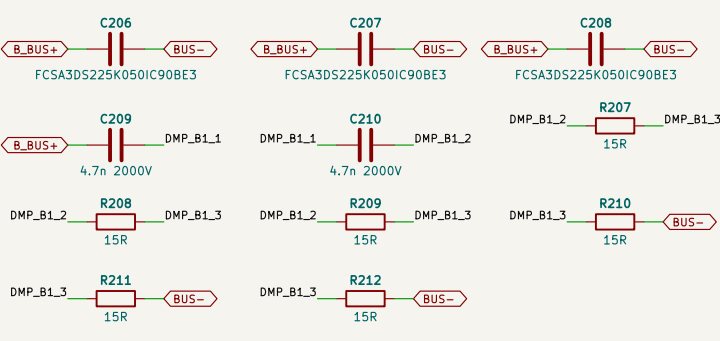
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



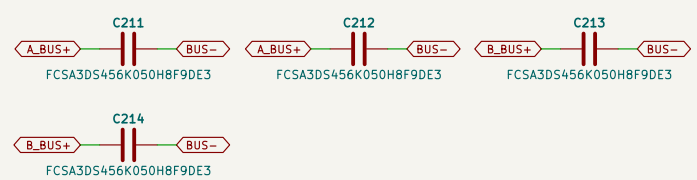
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



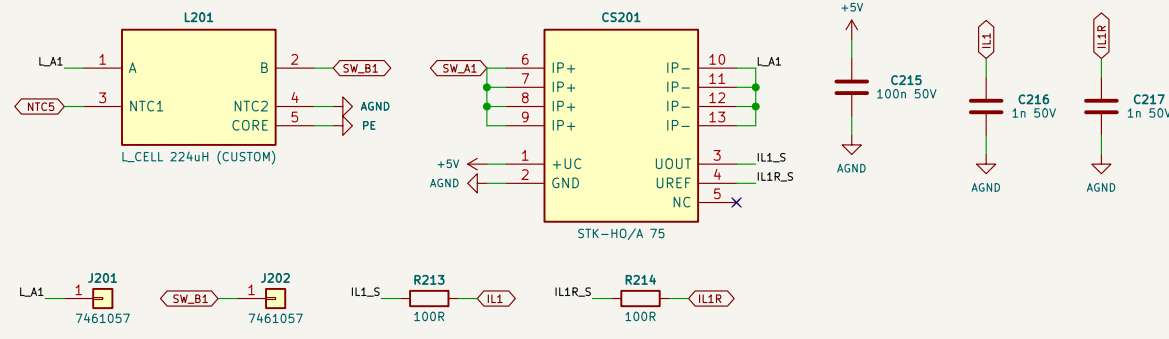
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



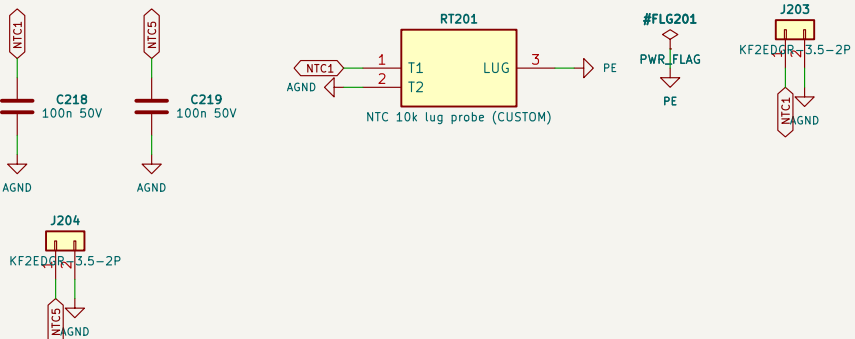
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



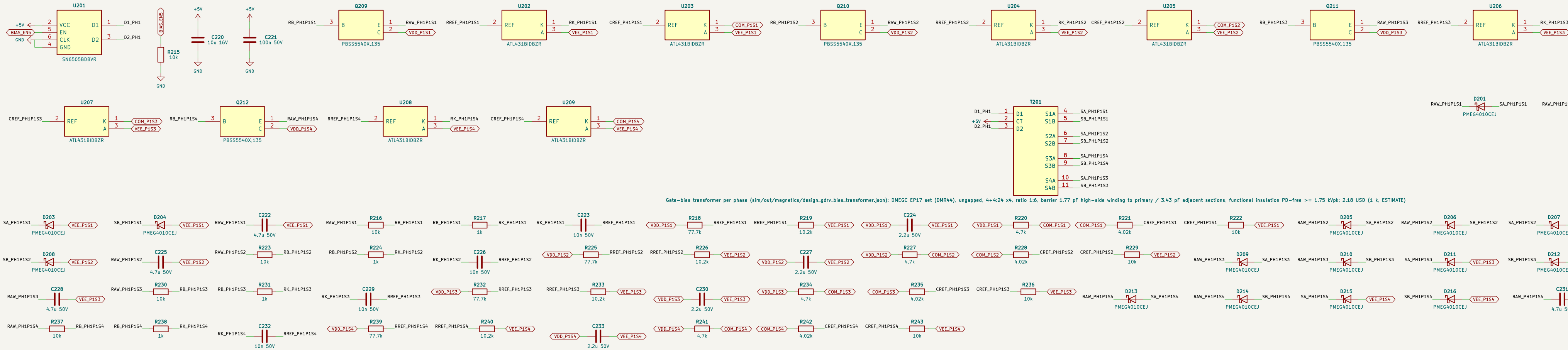
Lead A from SW.A1 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out + = SW.A -> SW.B) to stud LA1: lead B on stud SW.B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A: 100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



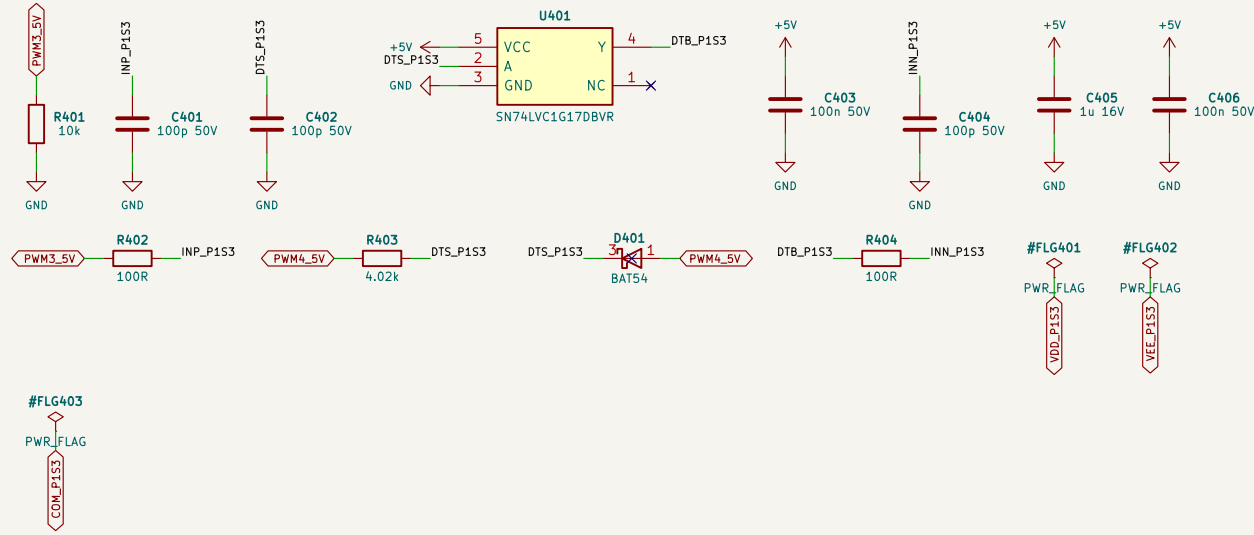
NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



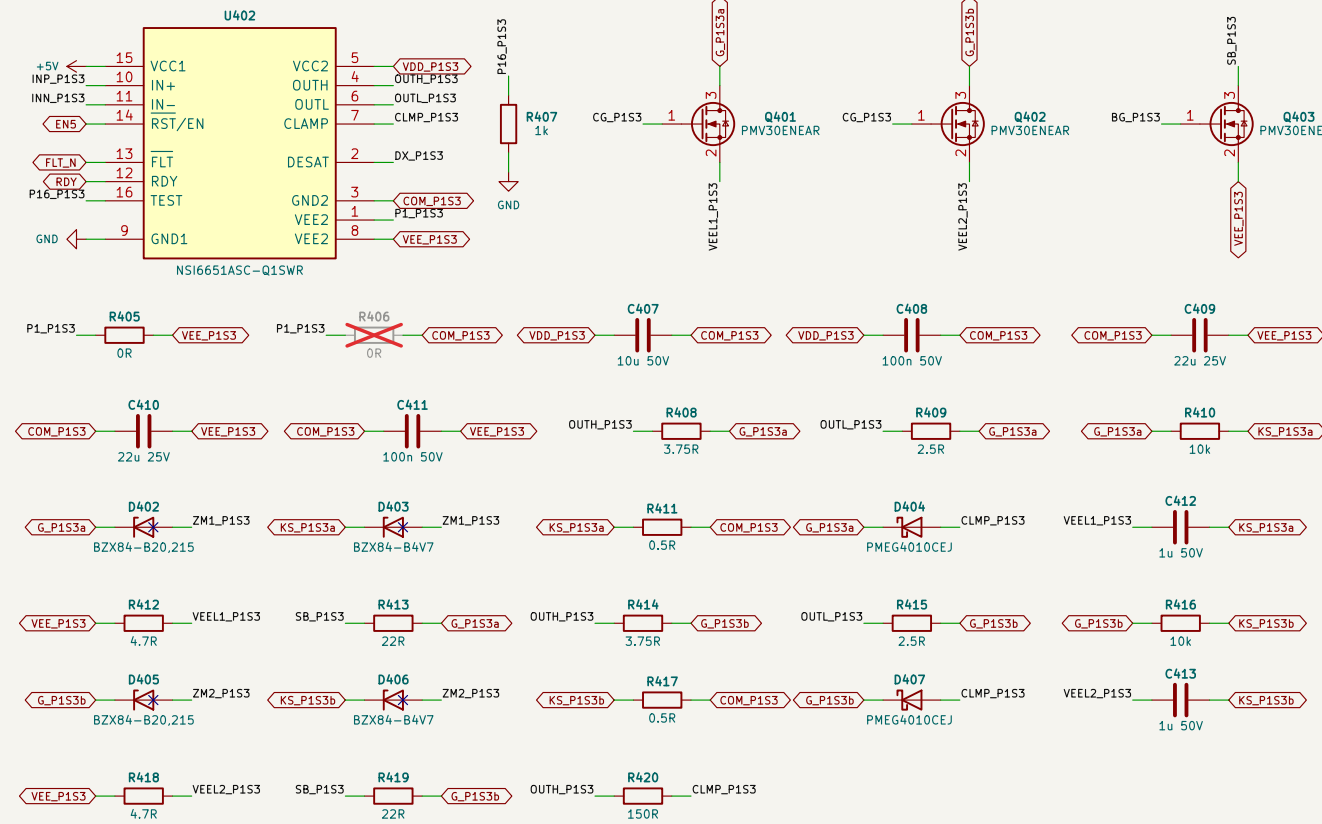
SN6505B push-pull 420 kHz from +5V. BIAS_EN high = on: T_BIAS4 custom, functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S3: LOGIC SIDE



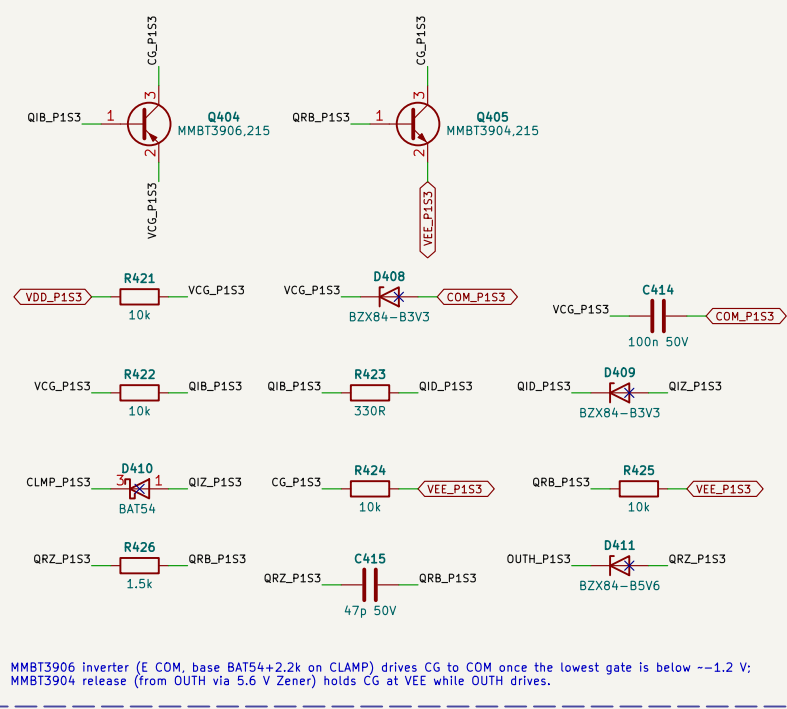
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



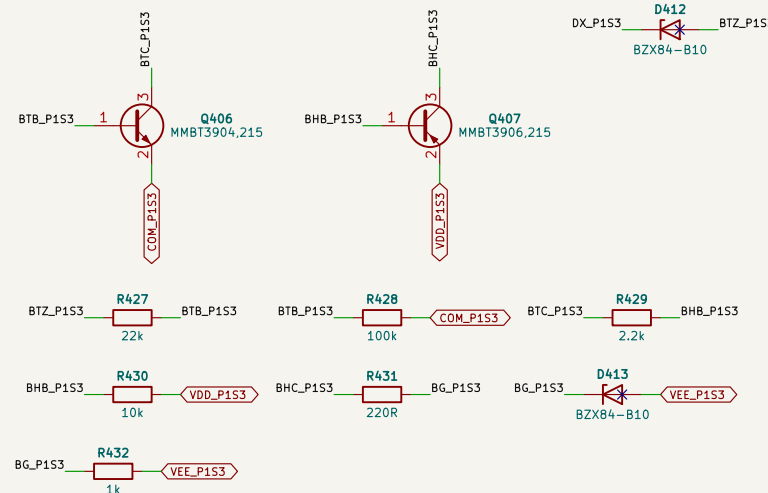
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



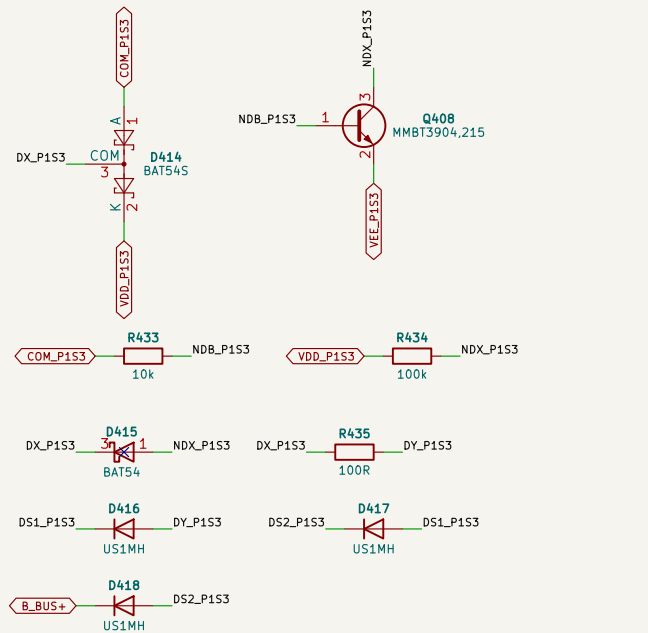
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



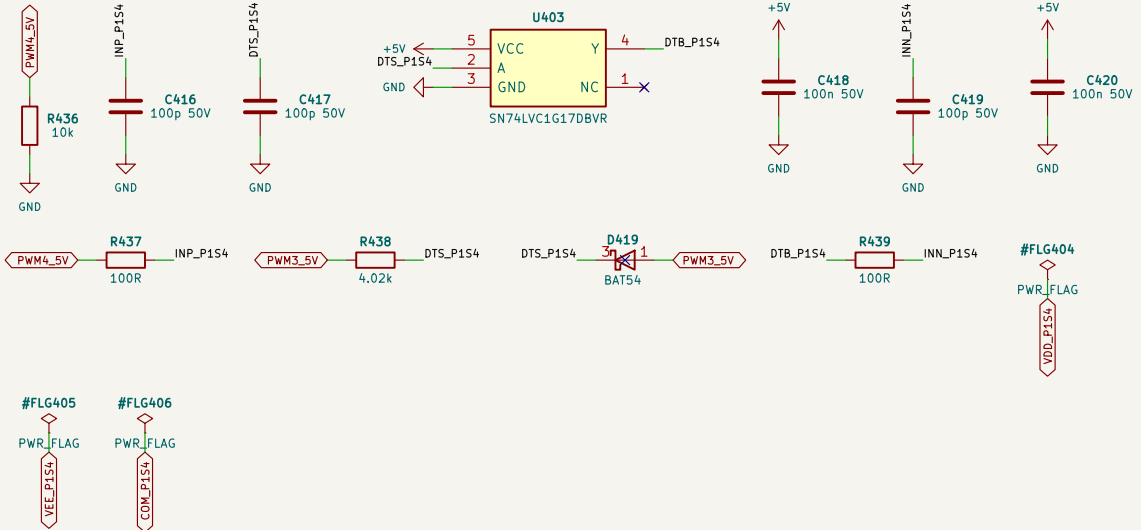
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S3: DESAT



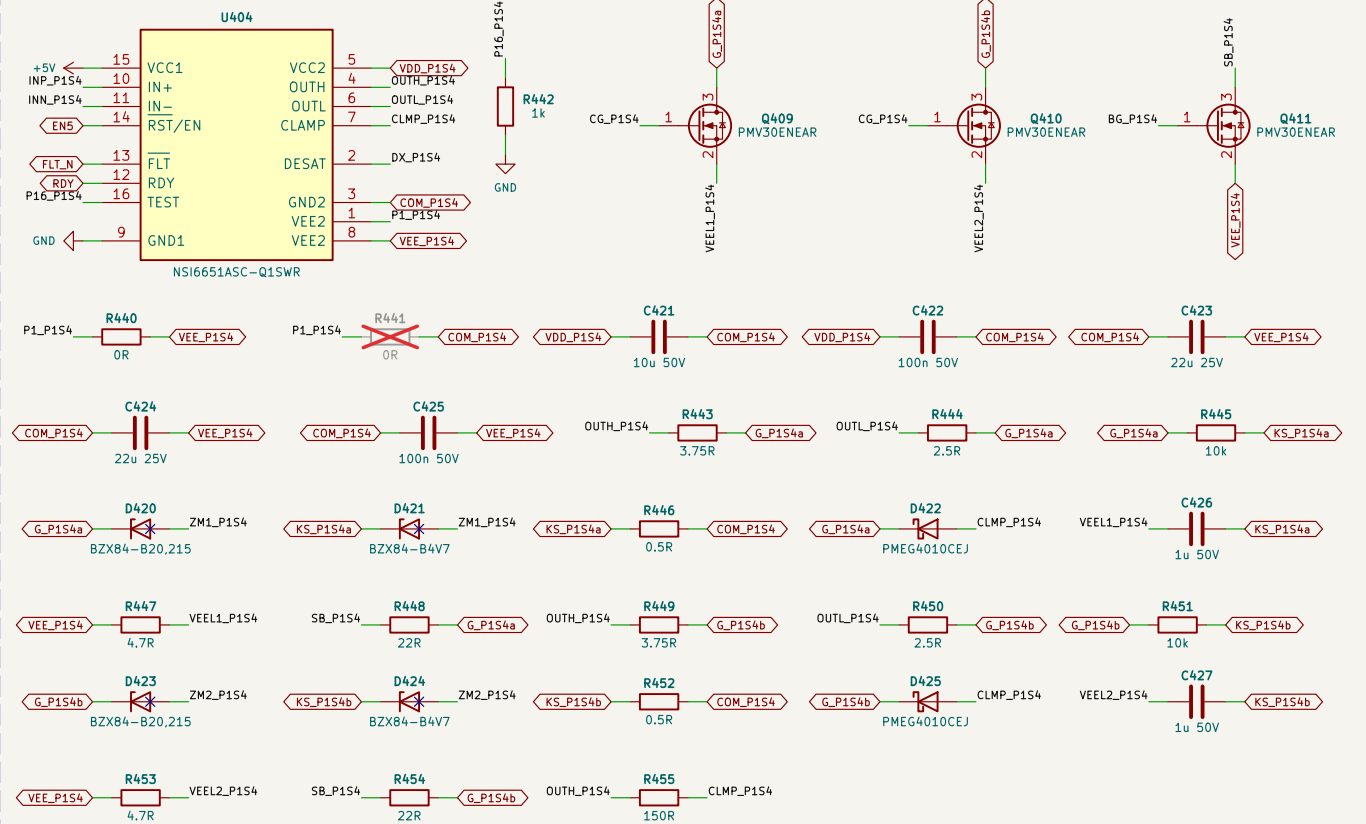
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S4: LOGIC SIDE



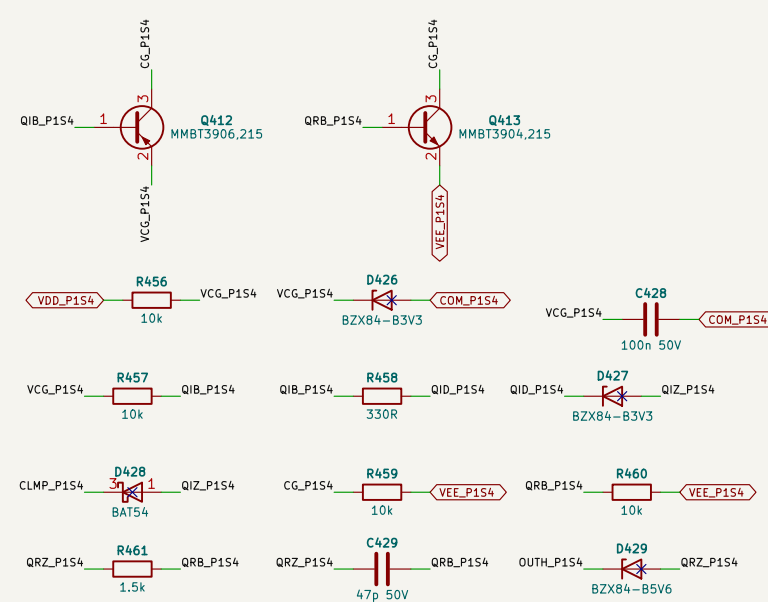
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



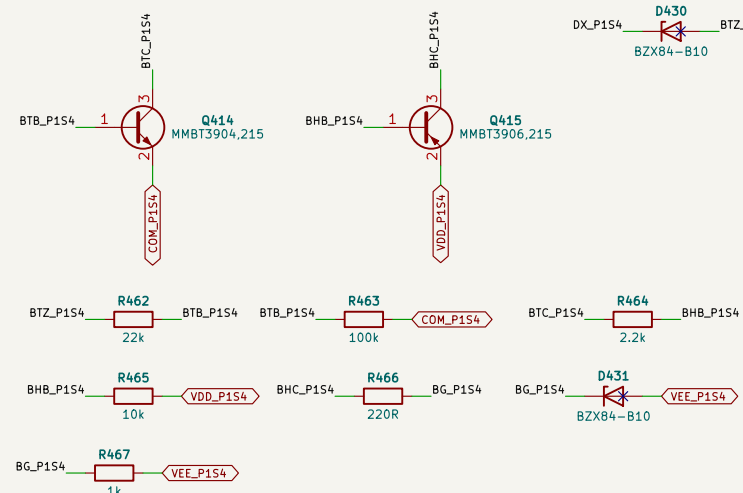
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S4: MILLER CLAMP CONTROL



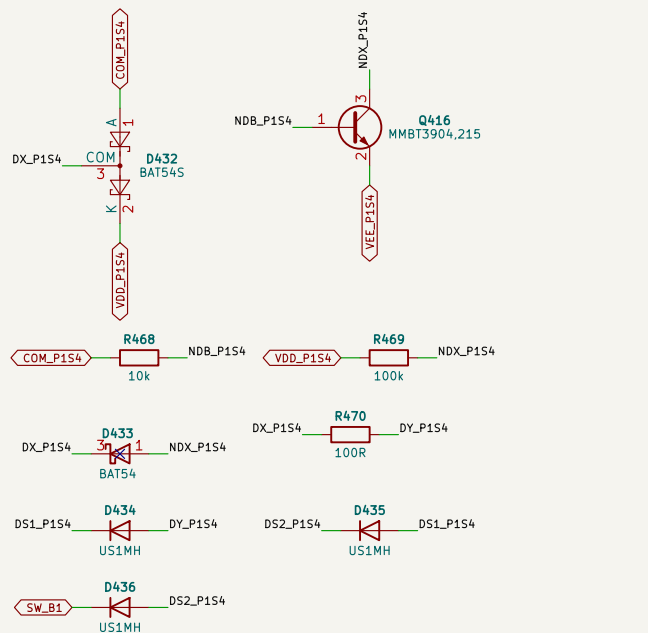
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

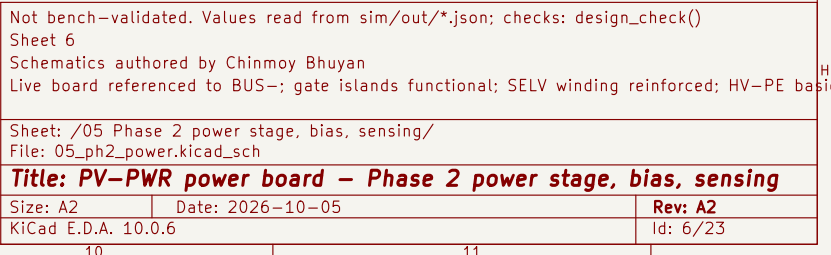
P1S4: DESAT



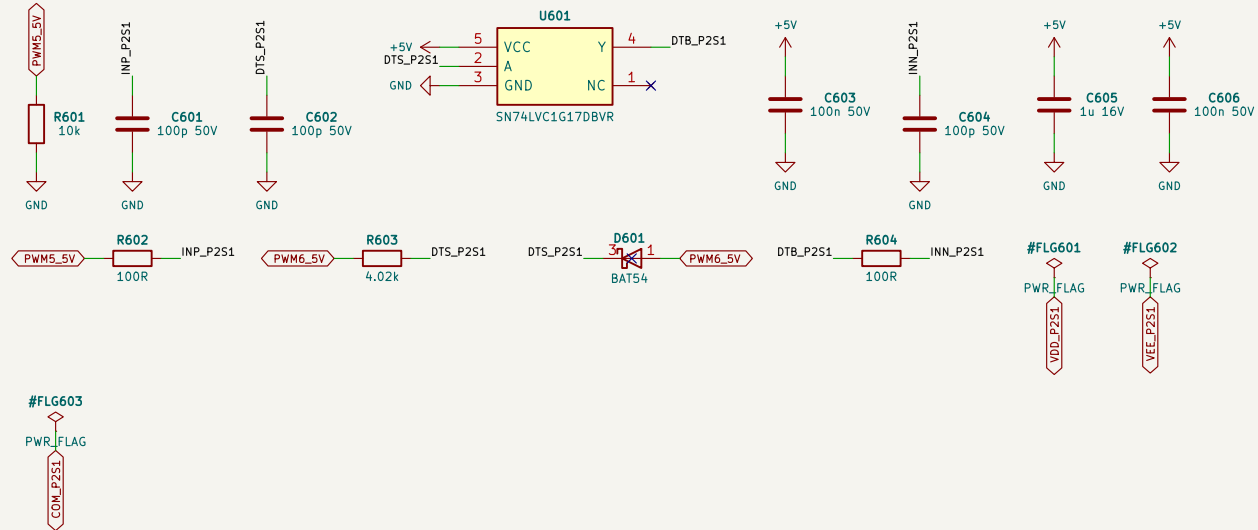
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

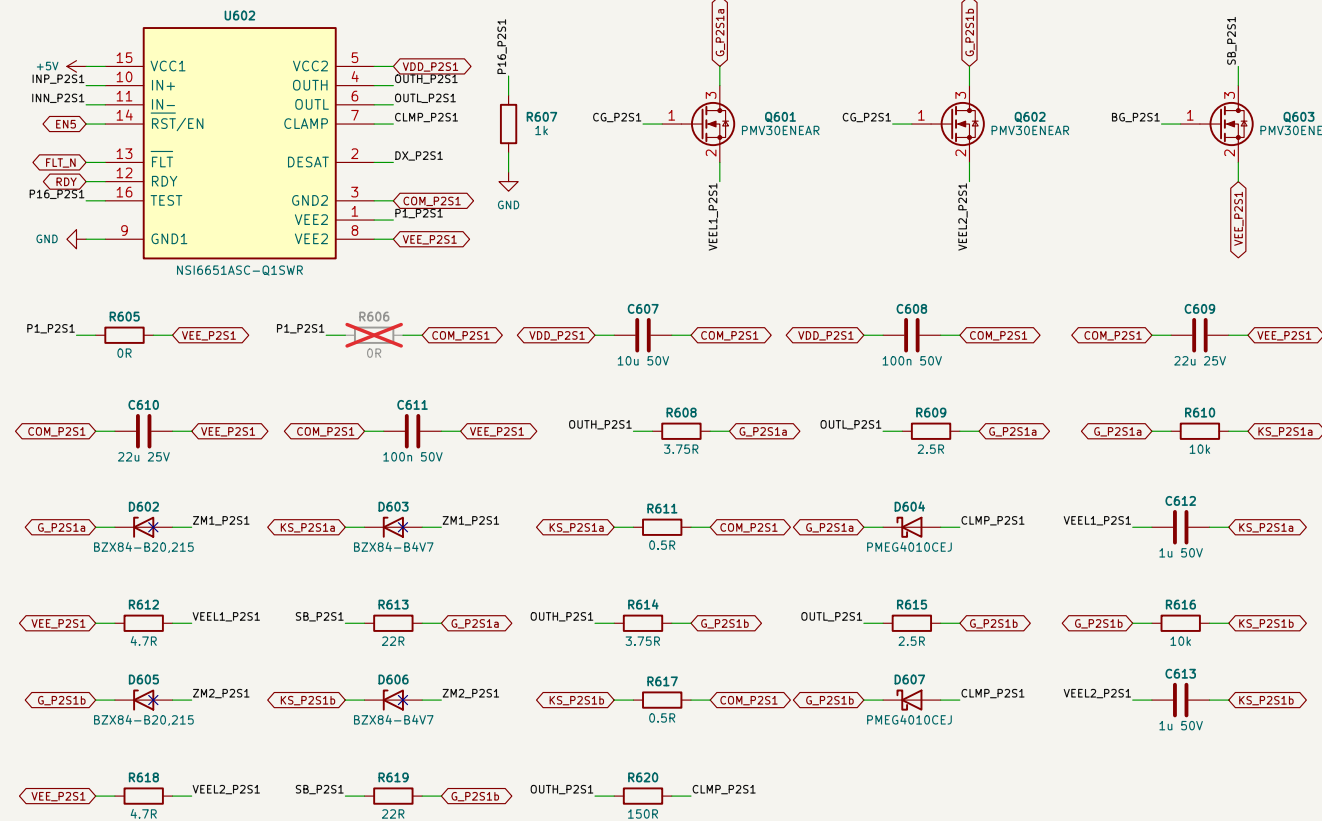


P2S1: LOGIC SIDE



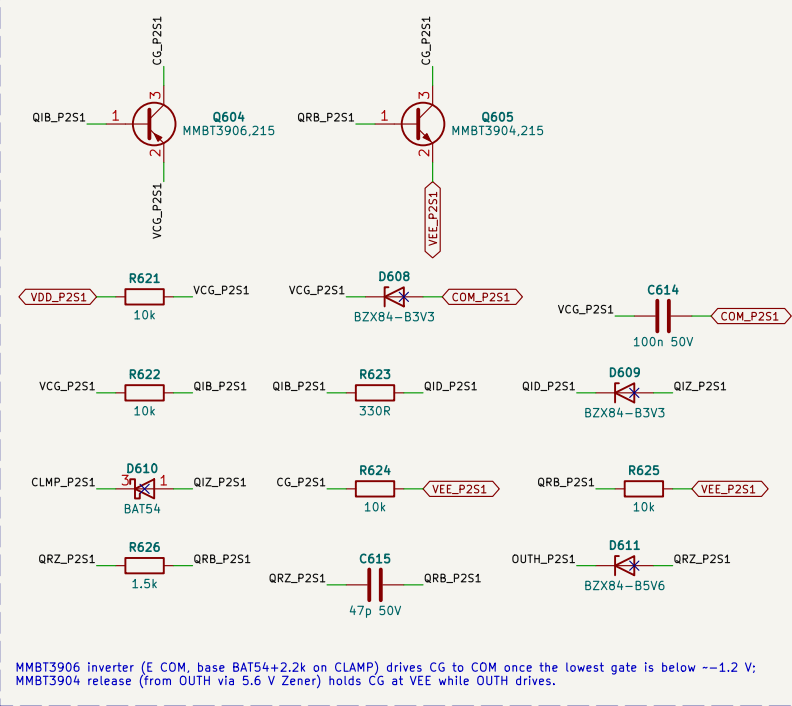
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



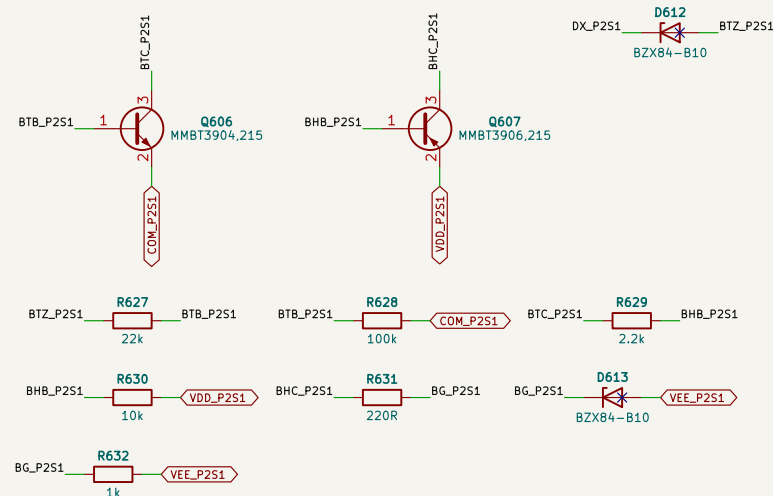
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMW30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL

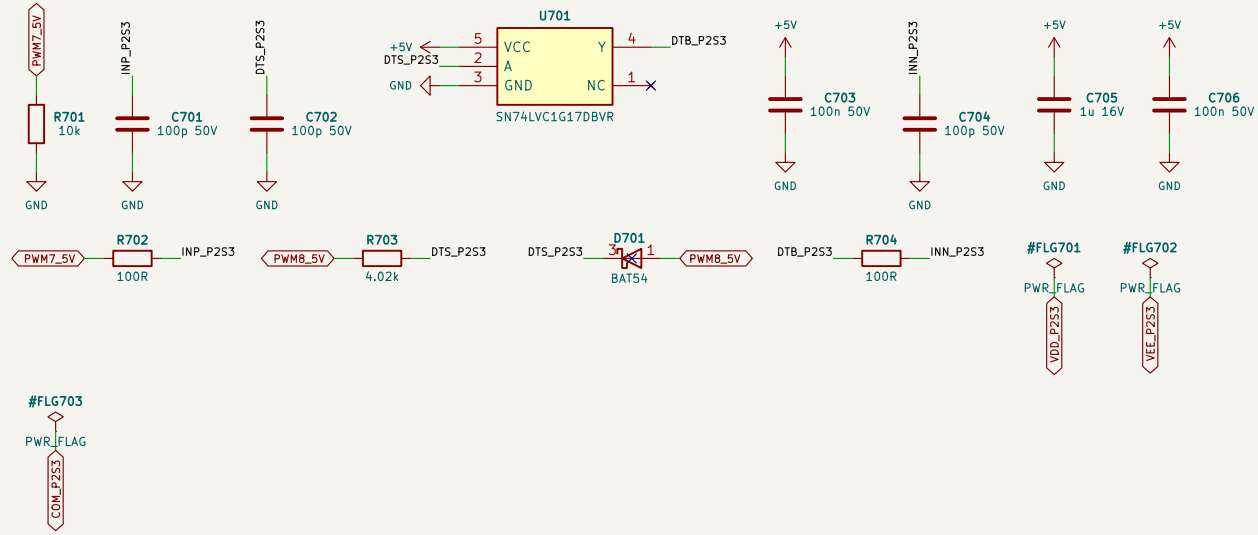


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER

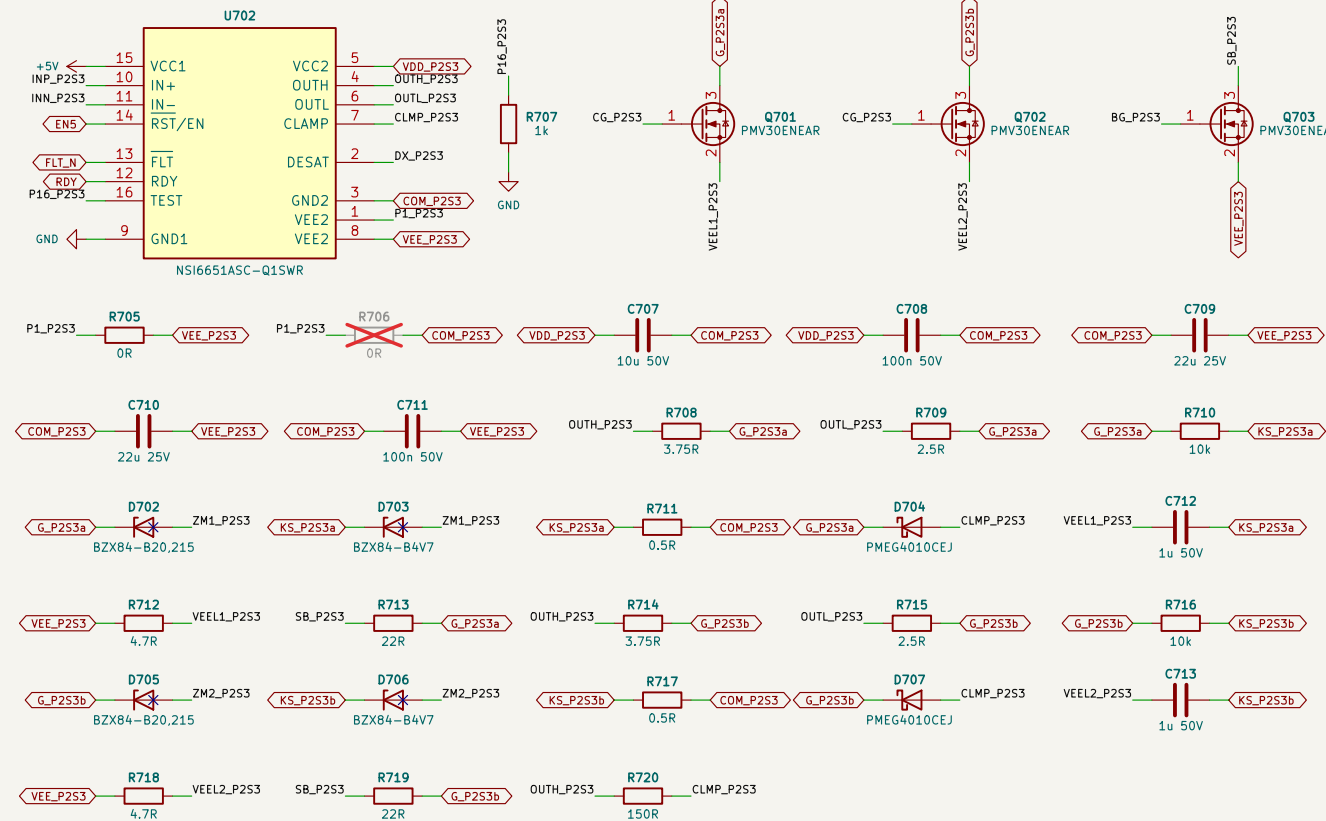


P2S3: LOGIC SIDE



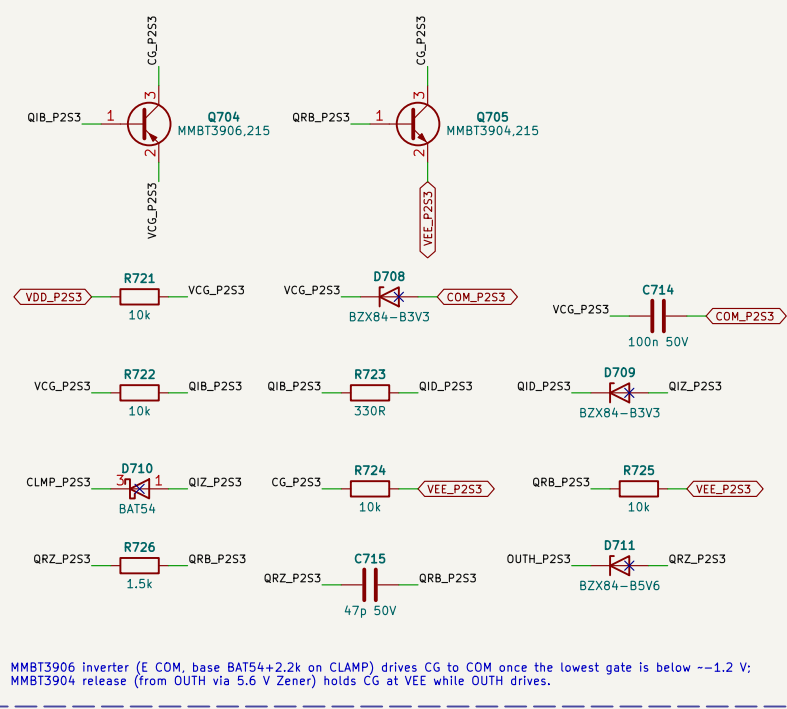
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



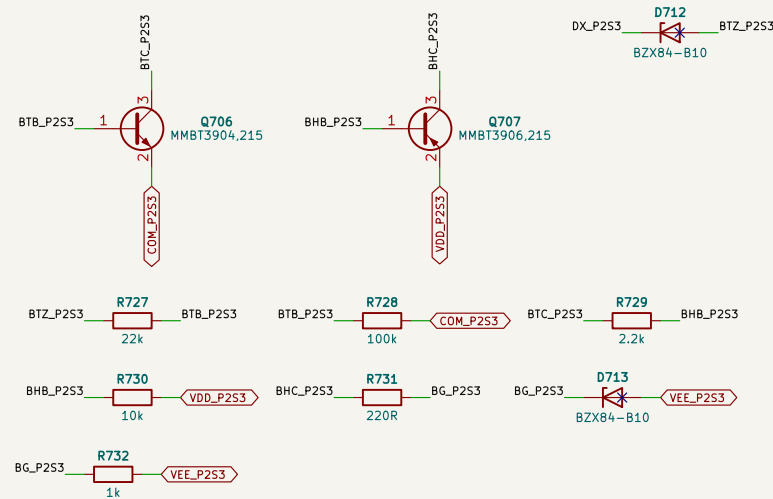
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL



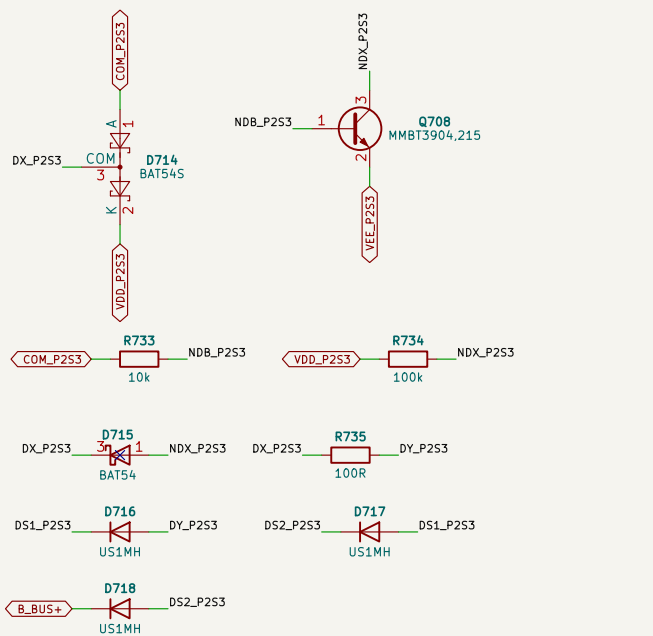
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER



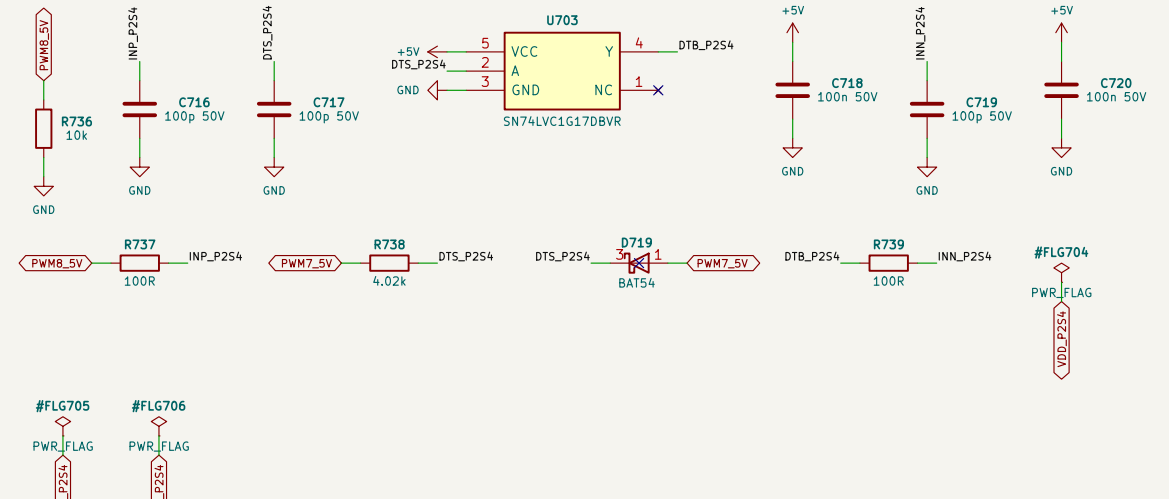
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P2S3: DESAT



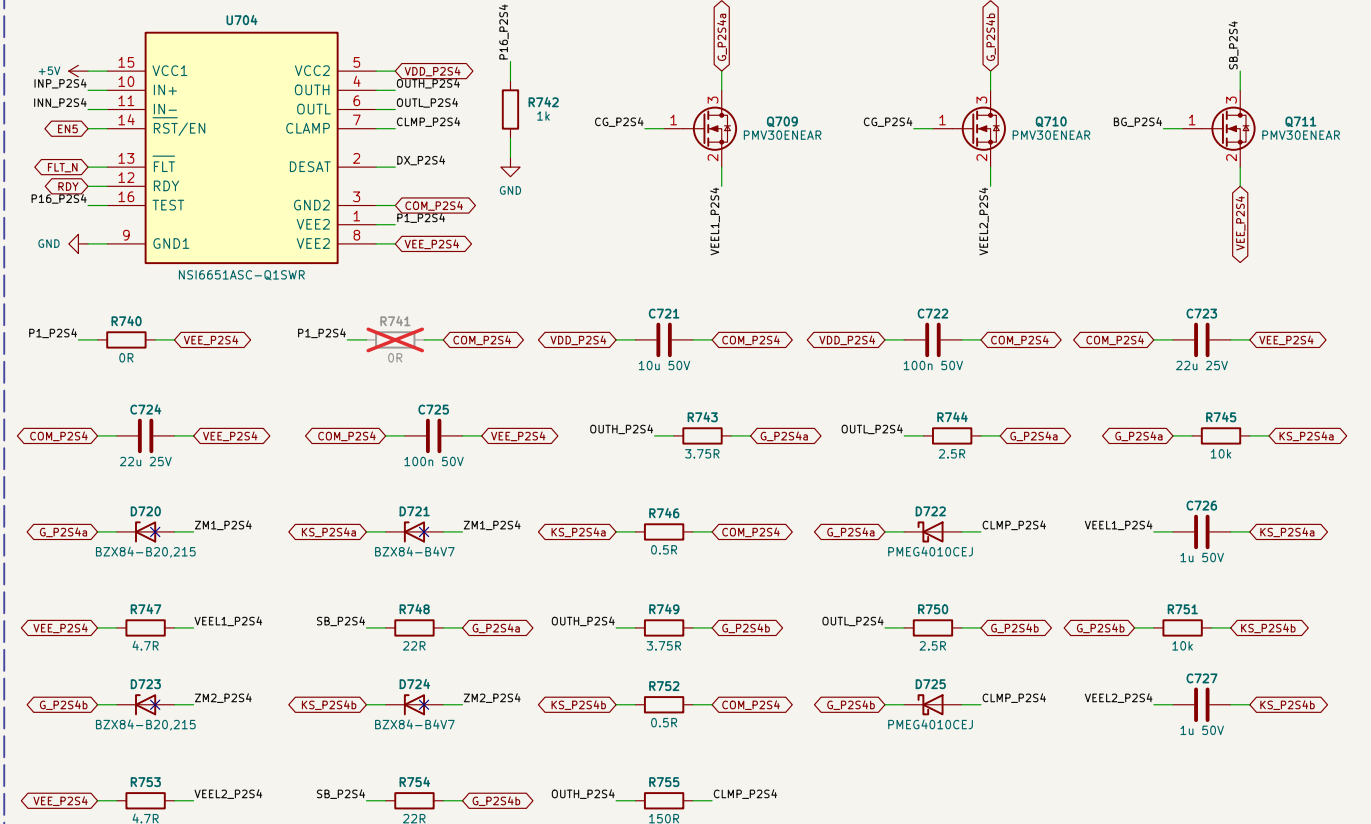
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S4: LOGIC SIDE



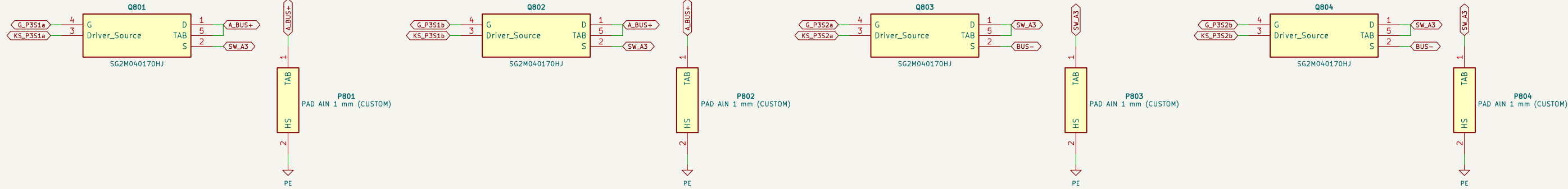
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S4: DRIVER AND 2 GATES



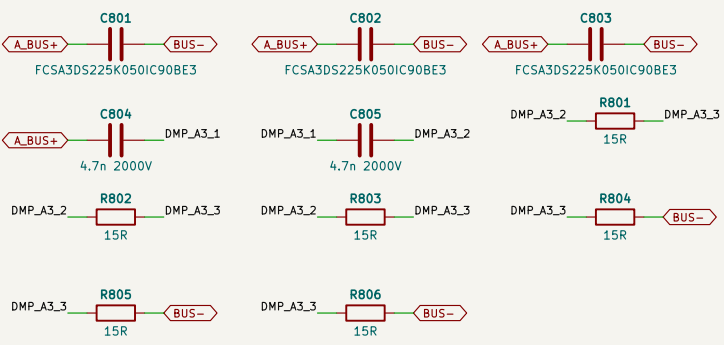
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



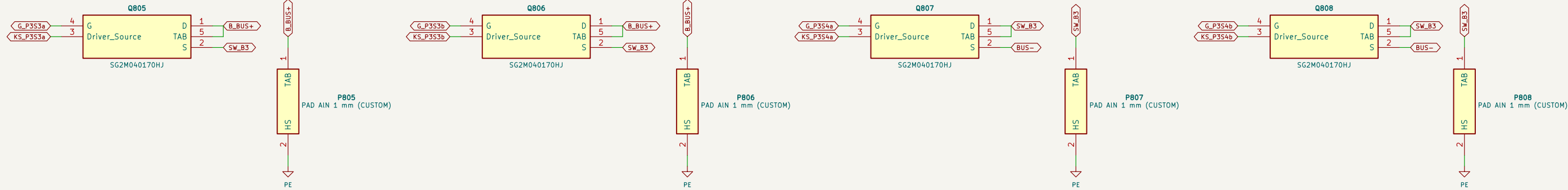
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



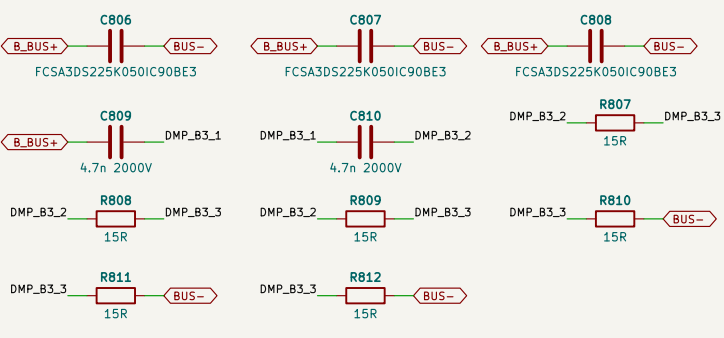
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



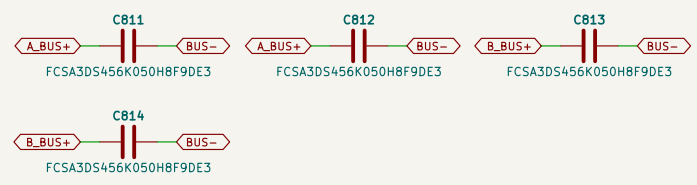
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



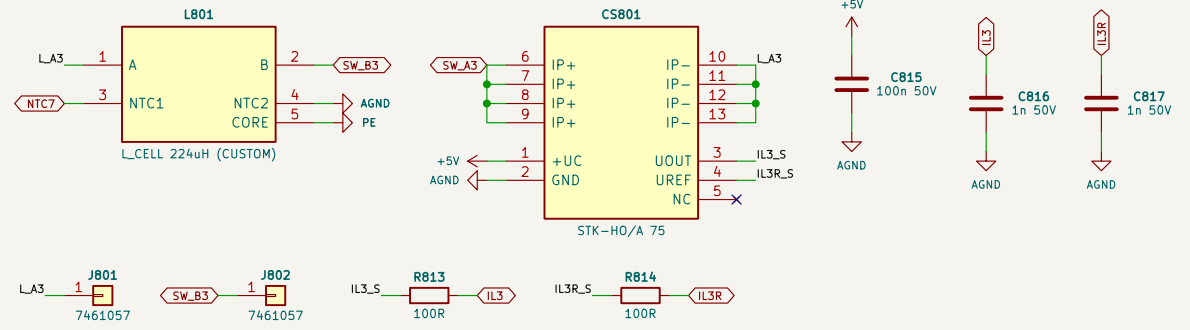
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



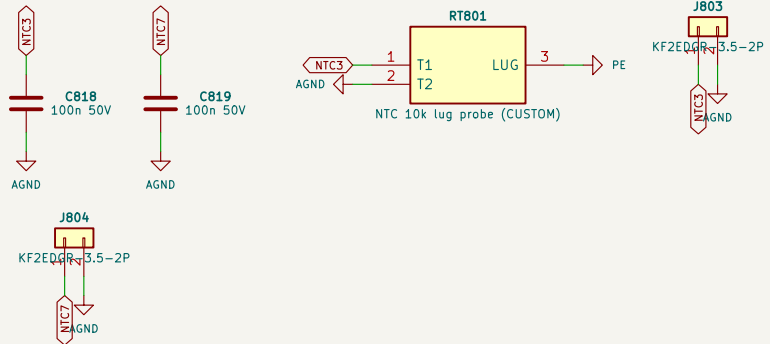
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



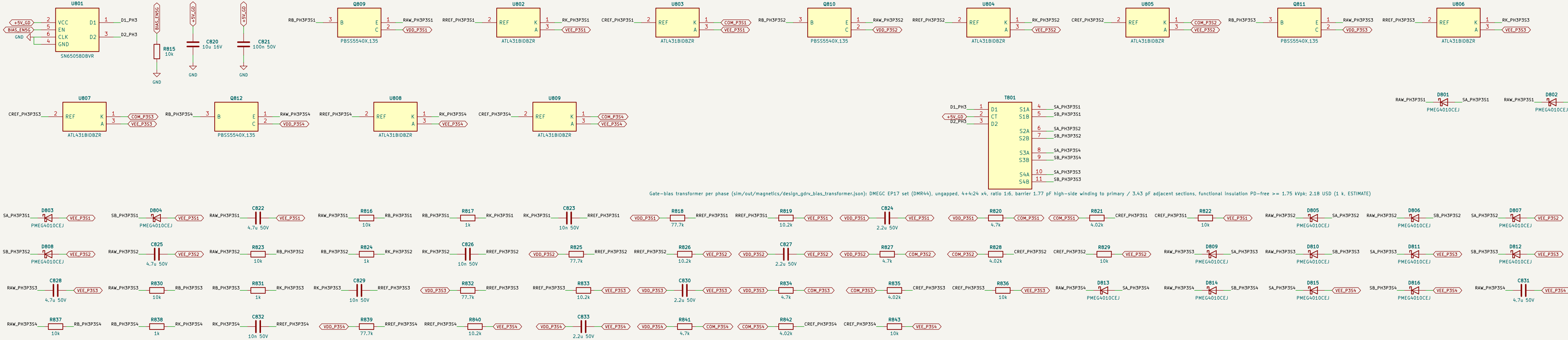
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out) + = SW_A -> SW_B to stud
LA3: lead B on stud SW_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mV at the source. IL3R = the sensor's own Uref (pin 4). 100R +/- 1 mV: PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



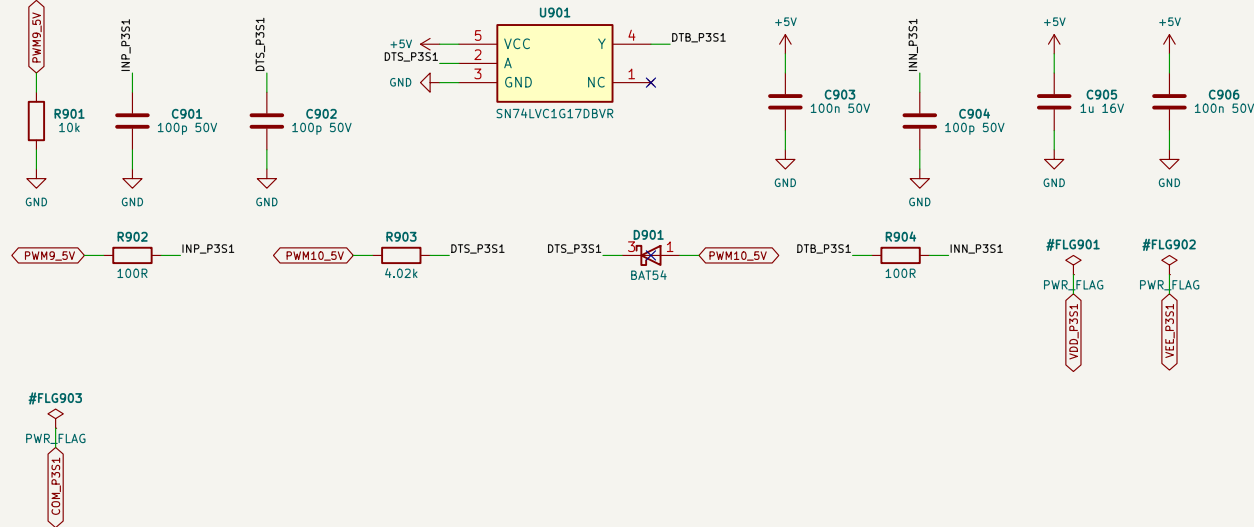
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC7 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



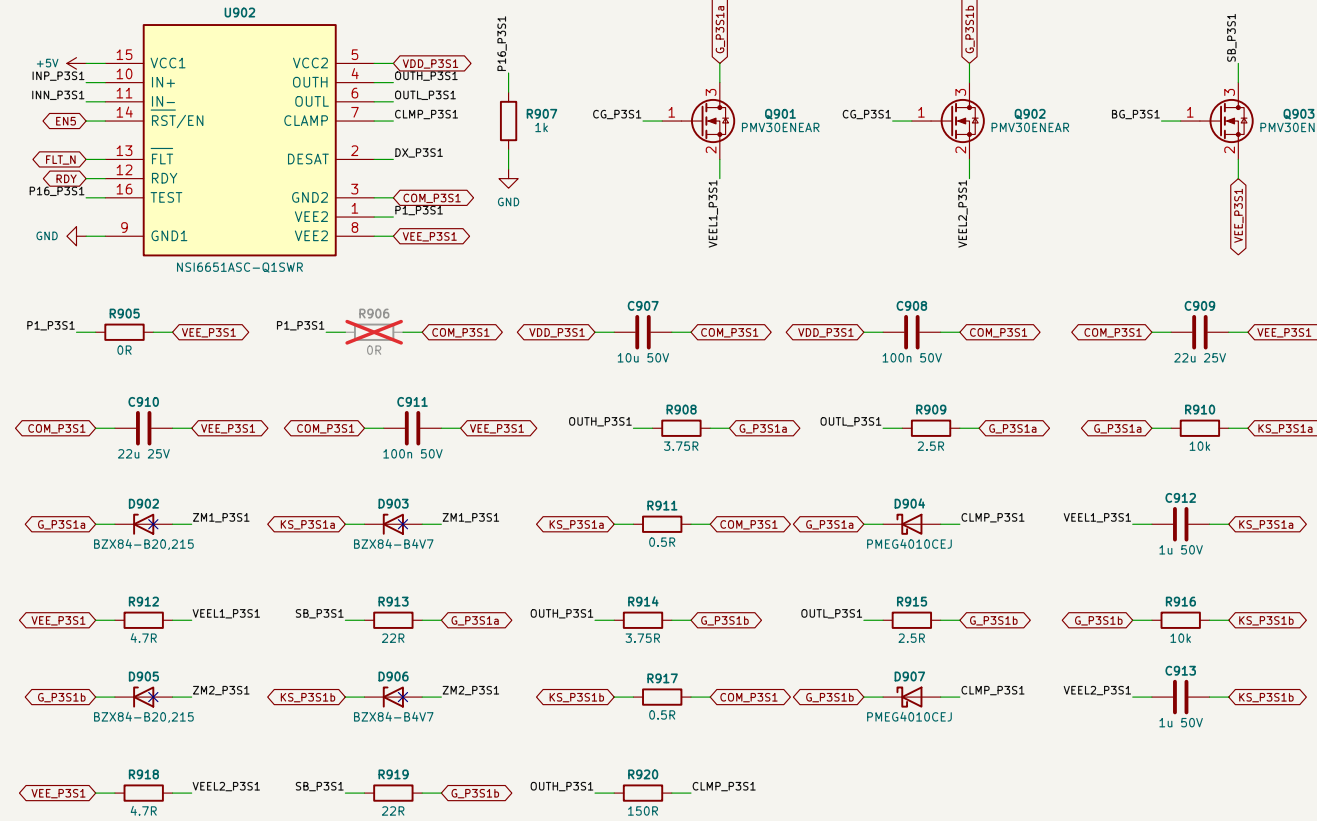
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T.BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



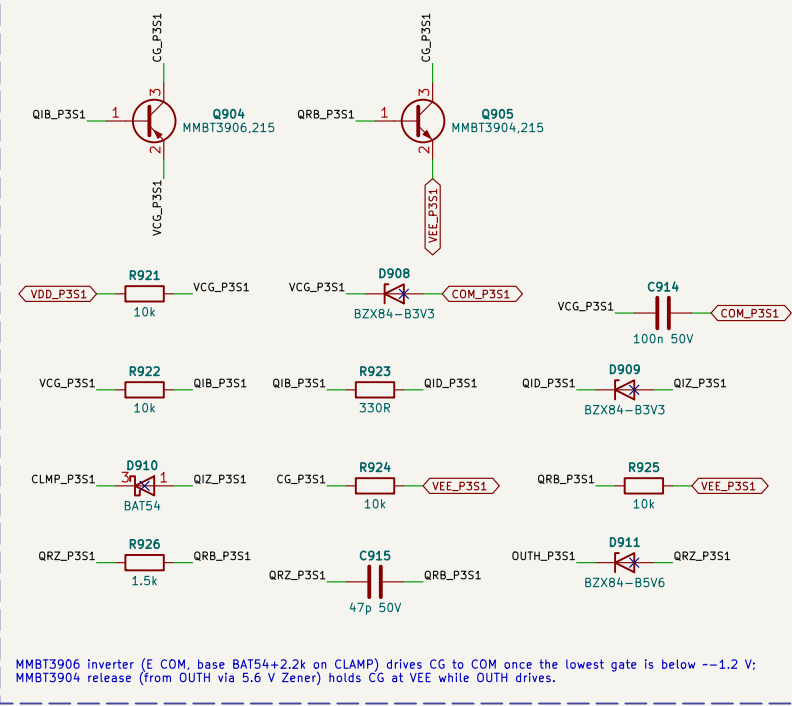
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



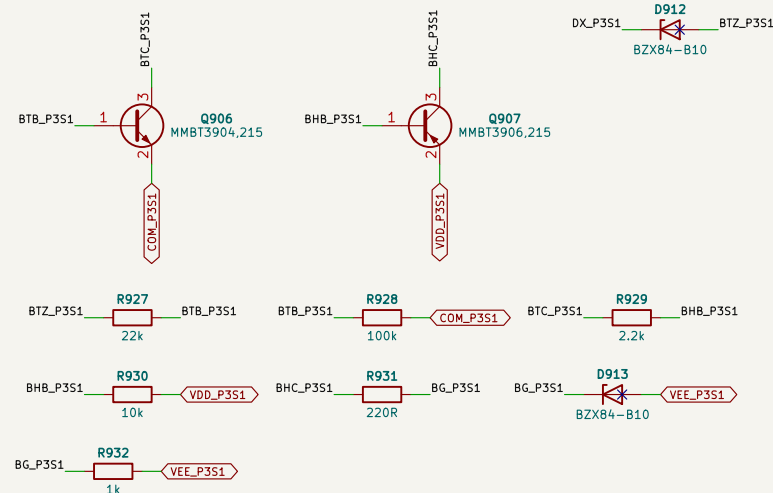
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL

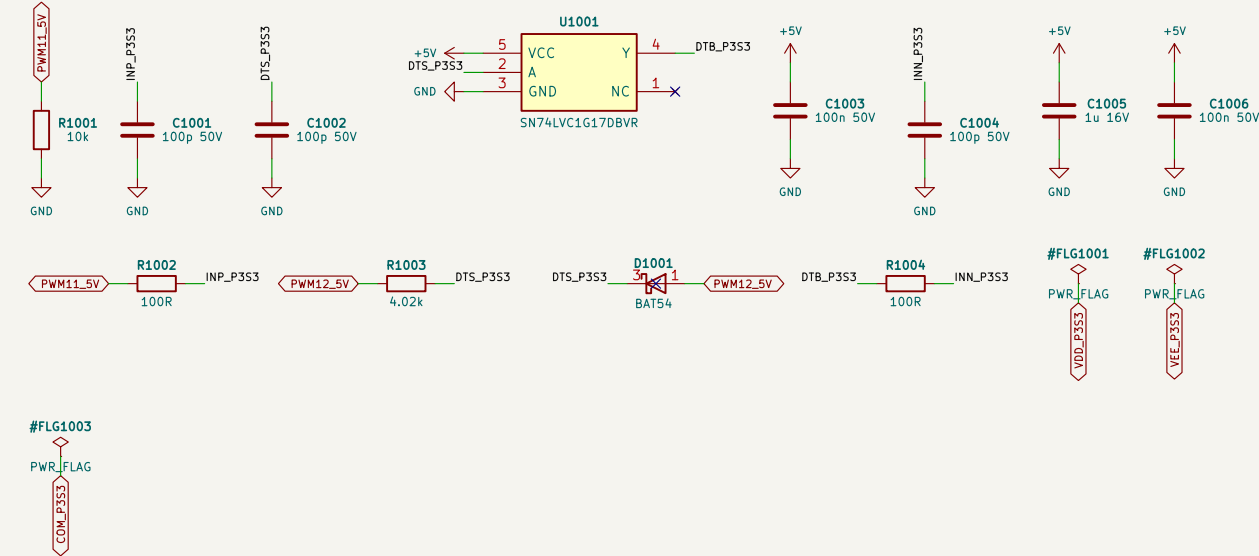


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER

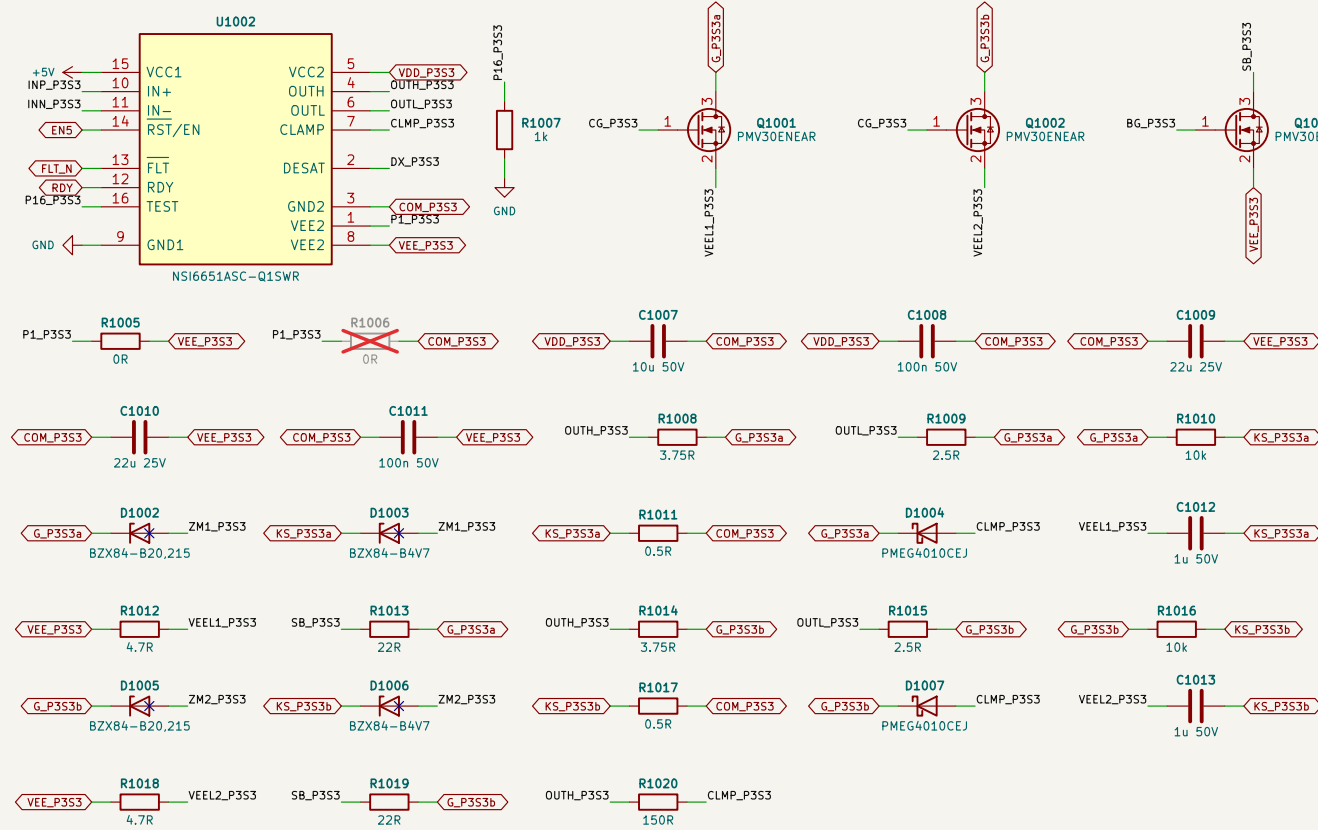


P3S3: LOGIC SIDE



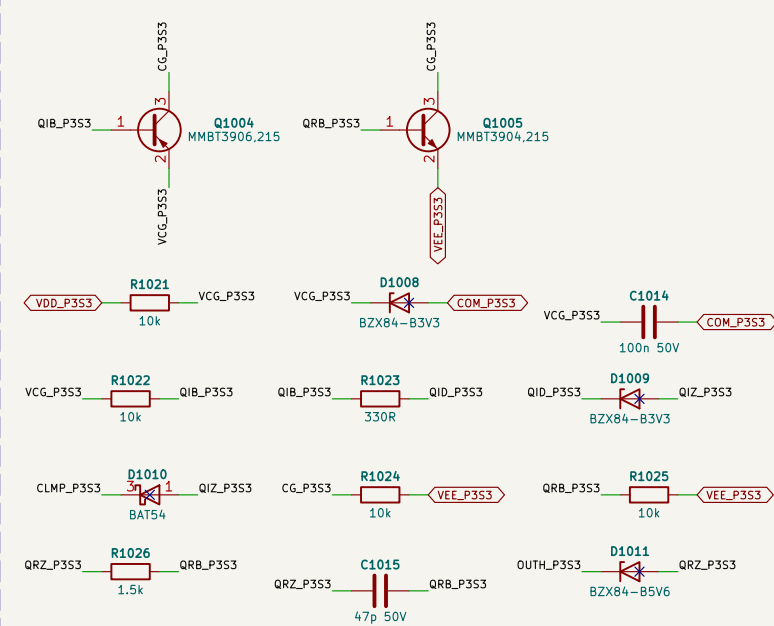
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



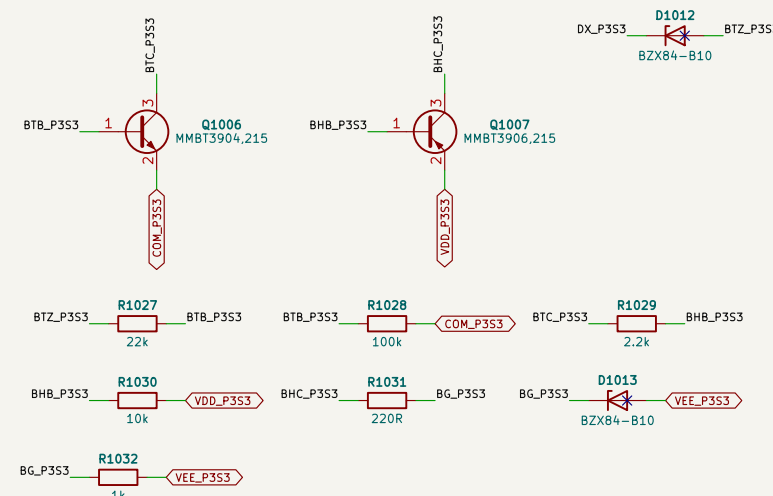
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



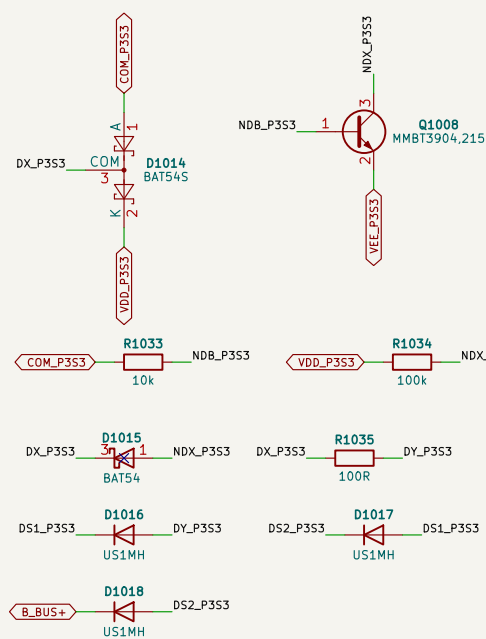
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



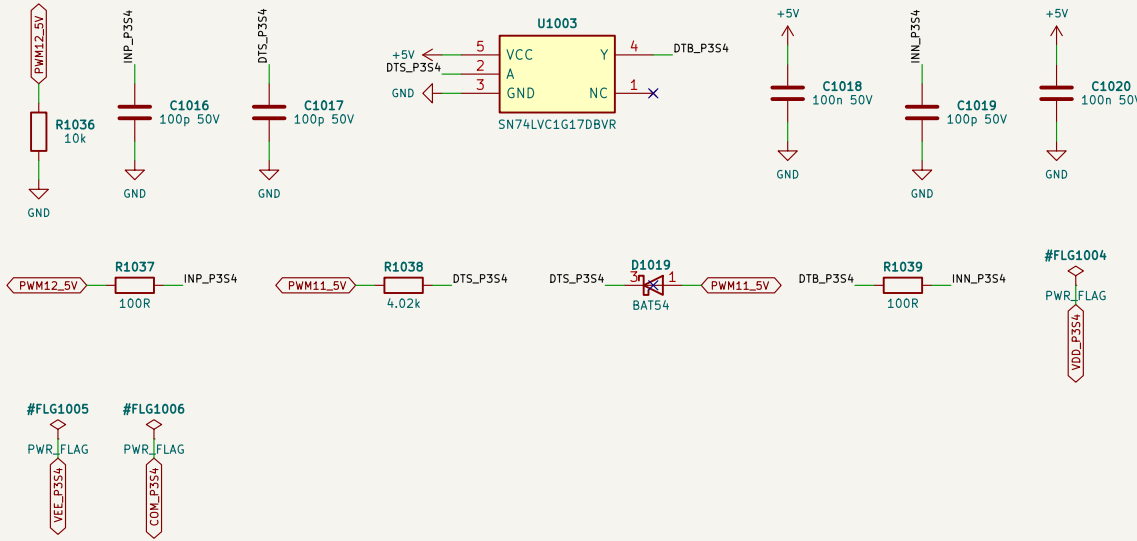
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S3: DESAT



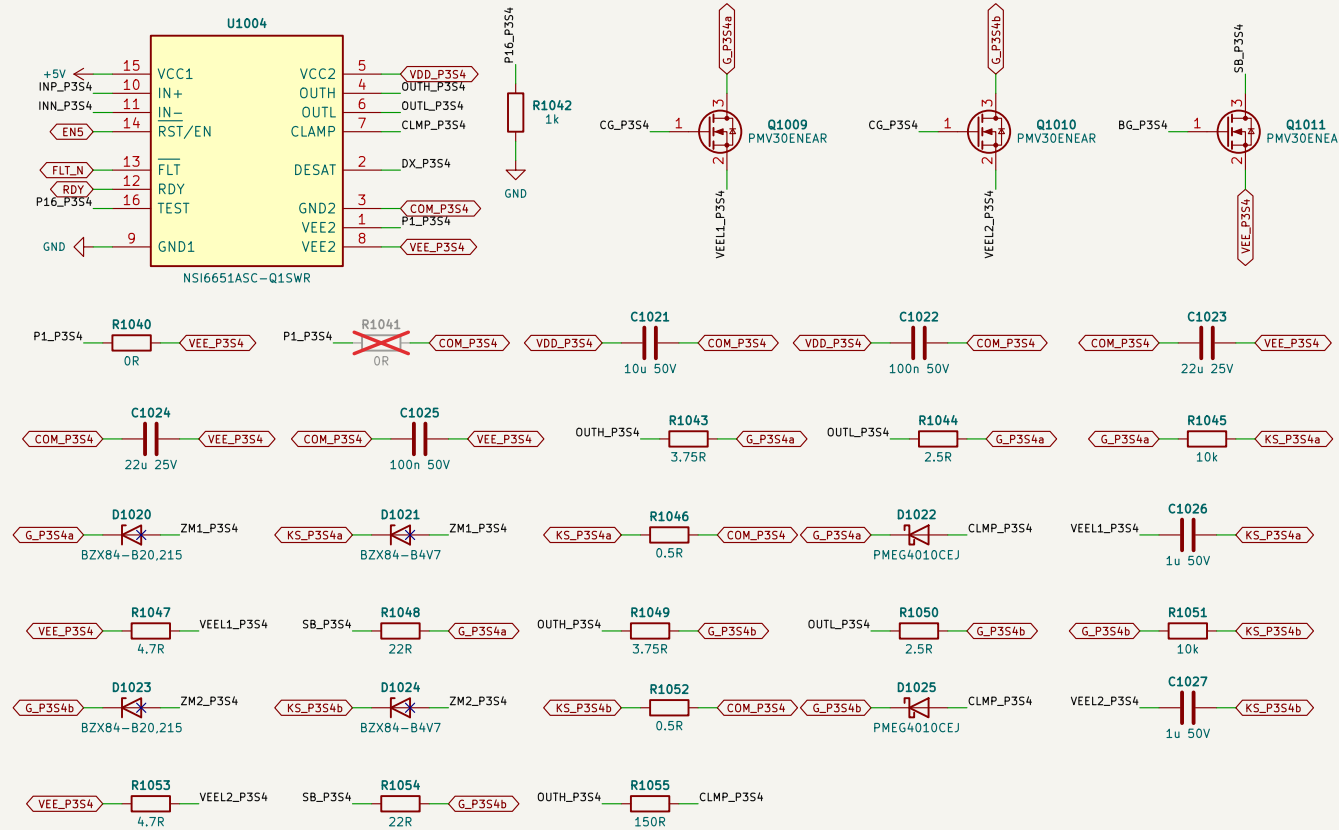
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S4: LOGIC SIDE



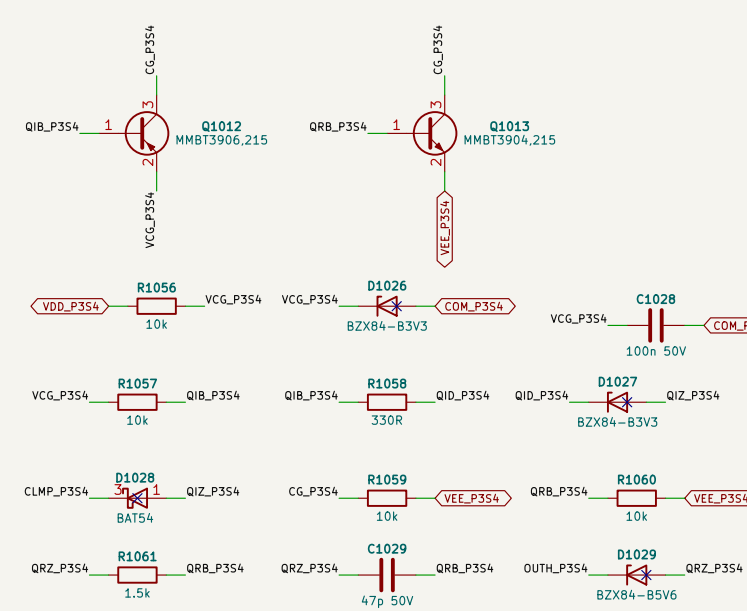
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



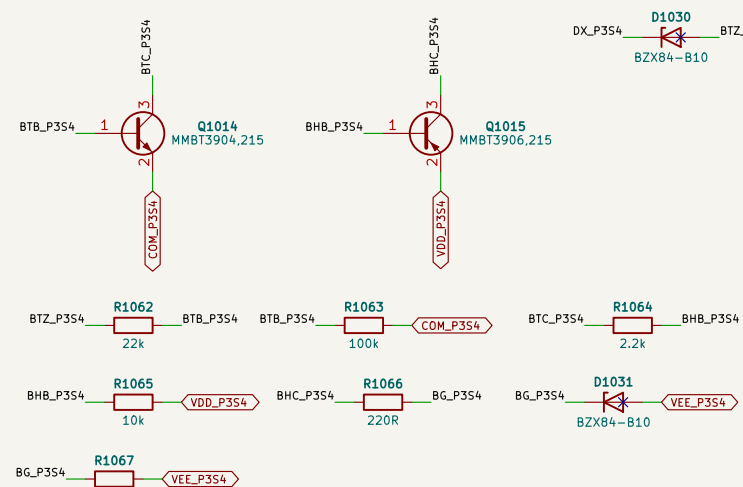
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



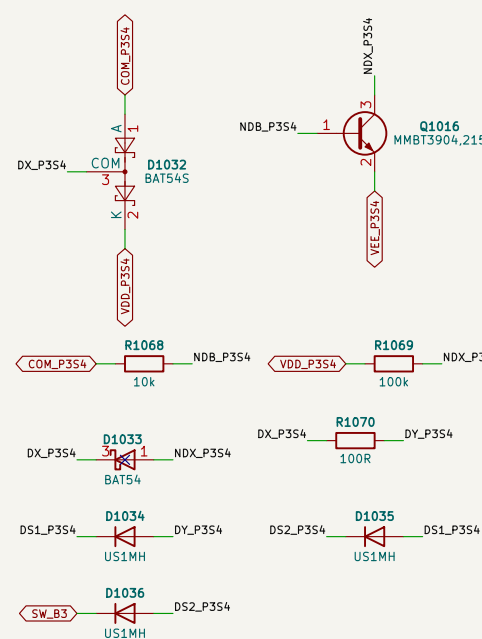
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 °C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /10 Phase 3 leg B gate drive (S3, S4)/

Title: PV-PWR power board – Phase 3 leg B gate drive (S3, S4)

Size: A2	Date: 2026-10-05
----------	------------------

Rev: A2

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array I_{sc} ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T, Up,eff 3793 V pole-P and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); Imax 25 kA, Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Not bench-validated. Uses read from sim/out/*.json; checks: design_check() Sheet 12		
Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands insulating; SELV winding reinforced; HV-PE basis		
Sheet: /11 Port A power path (pv)/ File: 11_leanA_power.kicad_sch		
Title: PV-PWR power board – Port A power path (pv)		
Size: A2 KiCad E.D.A. 10.0.6	Date: 2026-10-05	Rev: A2 Id: 12/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /12 Port A sensing and interlocks/
File: 12_leanA_ctrl.kicad_sch

Title: **PV-PWR power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 13/23

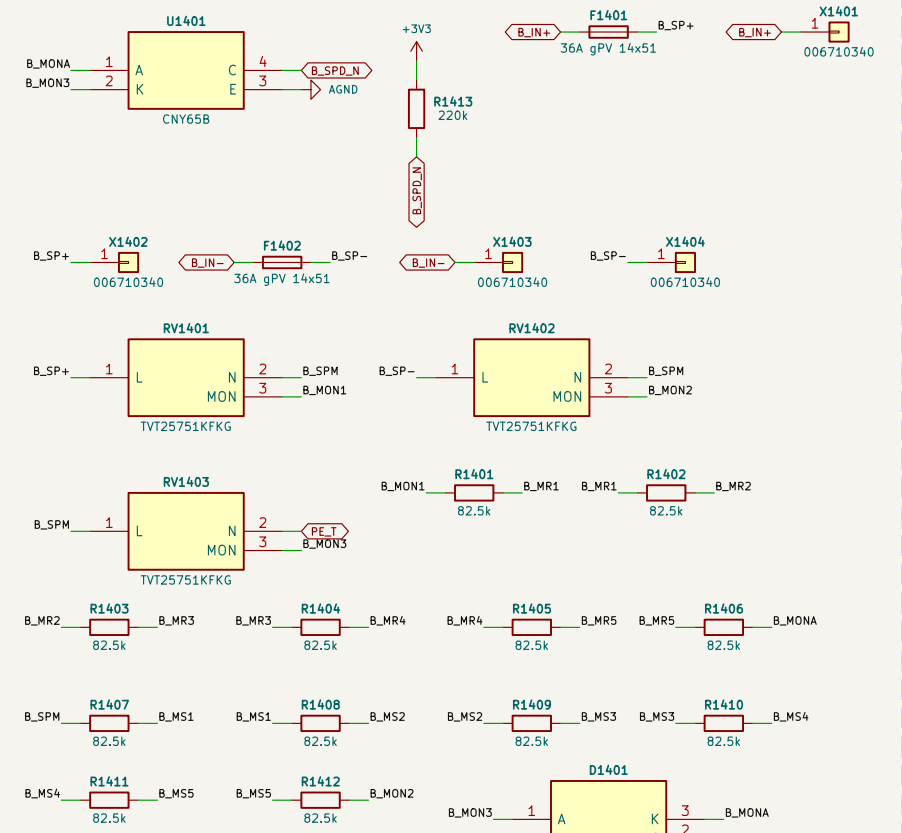
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 14 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /13 Port A coil drives/ File: 13_leanA_drive.kicad_sch		
Title: PV-PWR power board – Port A coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 14/23	

PORT B (BATTERY SOURCE), D-044 LEAN PORT

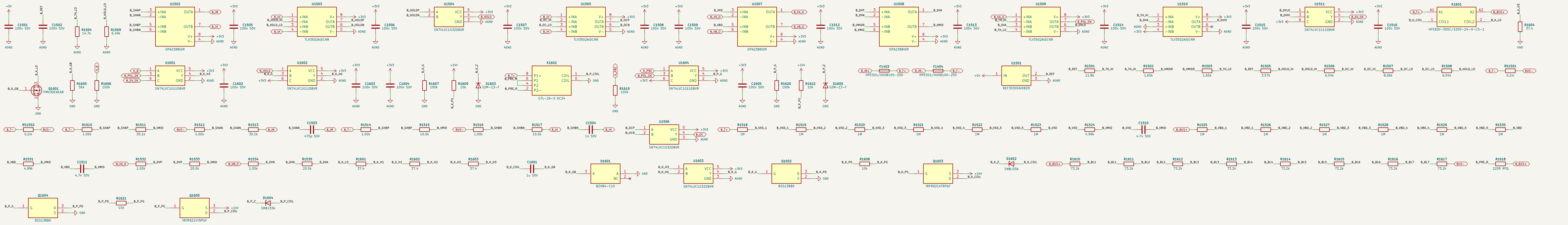
Terminal side R_MN+ / R_MN- bank R_BUS+/- HPE501 250 A at in both poles (breaks 1.25-50 A at in).
Contactor opens <= 0.97 A, held above (port_Lapc time). Installation: upstream circuiters 500-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV25751 x 3 (605 VDC, thermal time + monitor load); MOV1 +PM MOV2 -PM MOV3 W/PEL1 Use at 3795 V pole-PE and post-pole at 3 VA (GND, <= 4.0 VV => reinforced 5 VV / basic 4 V, D-032); max 25 mA Branch
From ESI CHANSEL GPF 50 A 50 mA (U/F 2 ms); Monitor loop through 500 Tares and all time (U/F => CHANSEL
-> R81 (sum on the ID line); R82 = 1 at rest above 200 V = varistor network open -> CTR1 inhibit/stop.

AR FUSES (BATTERY-TYPE PORT)



	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 16 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /14 Port B sensing and interlocks/ File: 14_leanB_ctrl.kicad_sch		
Title: PV-PWR power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 16/23	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Port B coil drives/
File: 15_leanB_drive.kicad_sch

Title: **PV-PWR power board – Port B coil drives**

Size: A3

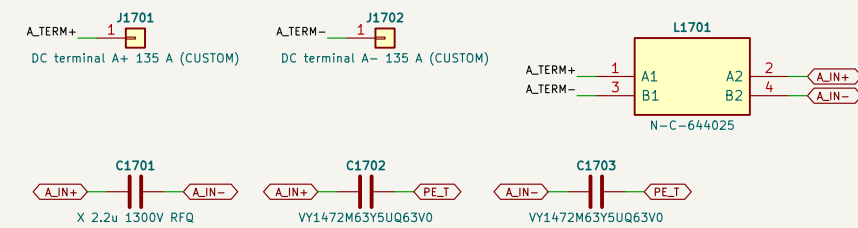
Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

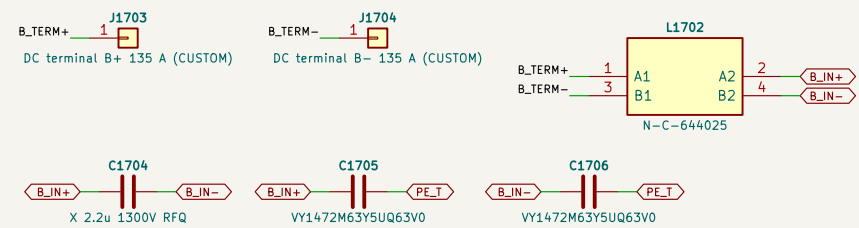
Id: 17/23

PORT A TERMINALS AND EMI FILTER



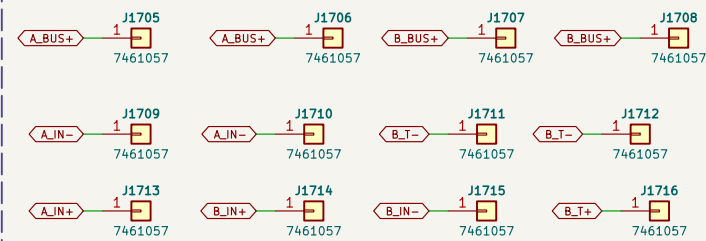
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board pads A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



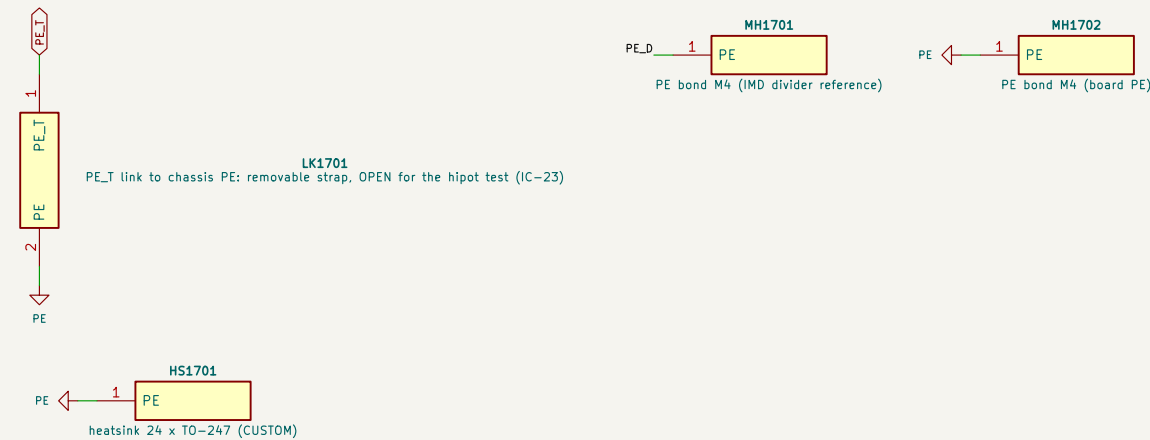
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



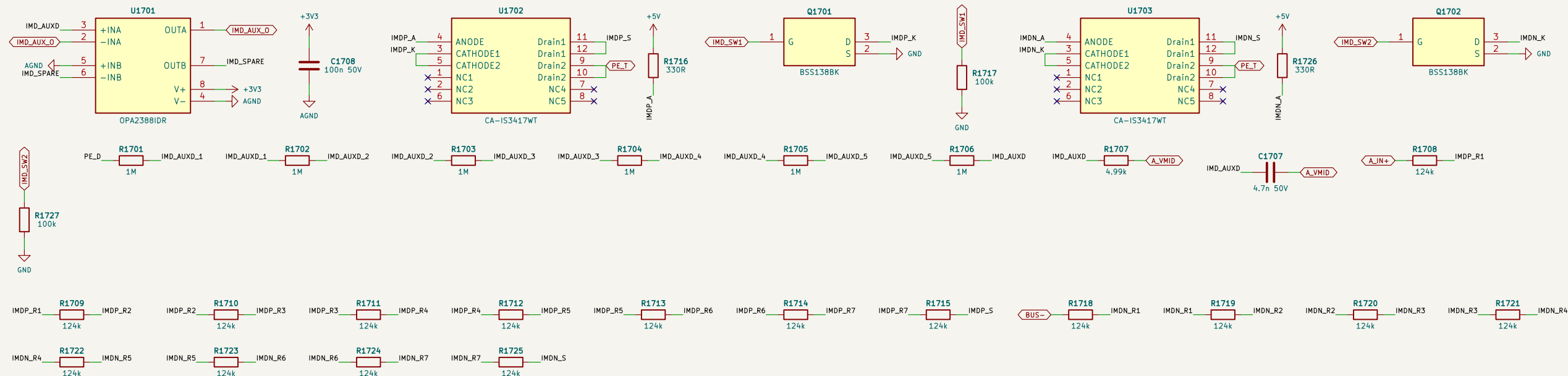
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

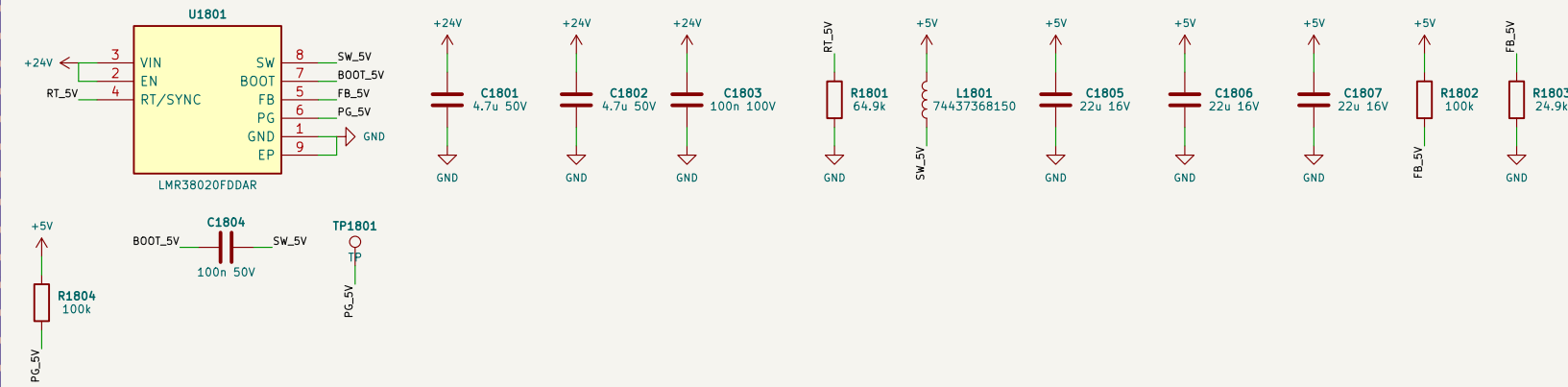
INSULATION MONITOR (PORT.LEAN_IMD)



```
IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected);
VPE = PE divider against BUS- (VMID of port A).
```

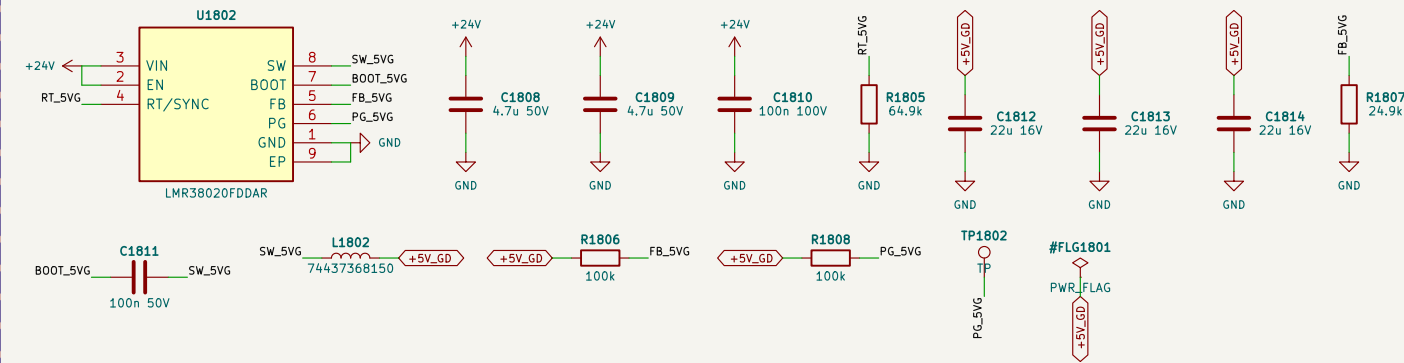
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 18 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Terminals, EMI filter, insulation test/ File: 15_terminals_imd.kicad_sch		
Title: PV-PWR power board – Terminals, EMI filter, insulation test		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 18/23	

24 V $\rightarrow$ +5V (LMR38020, 400 KHZ, 2 A)



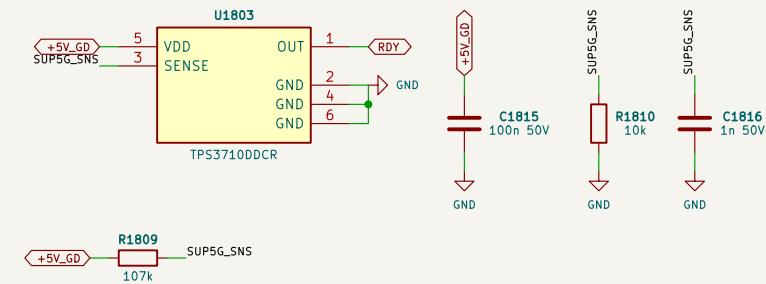
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2. ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



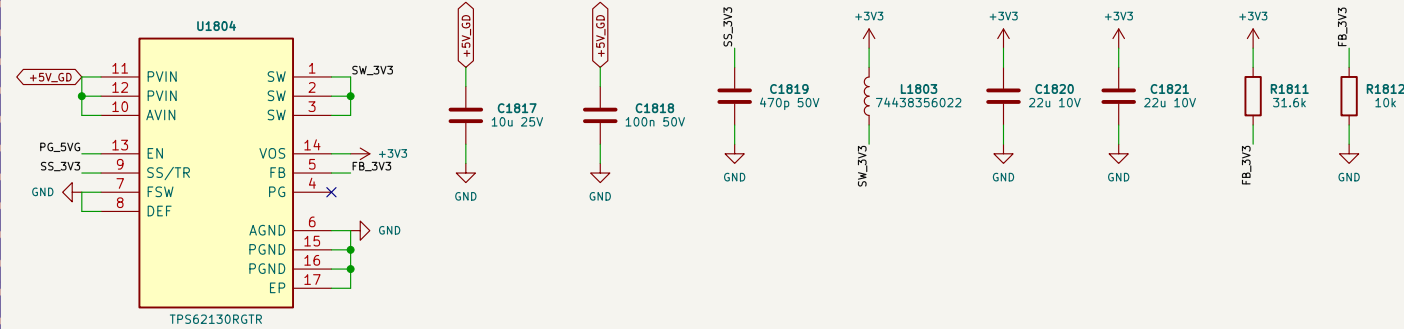
gate bias of phases 3 only. ctrlc2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



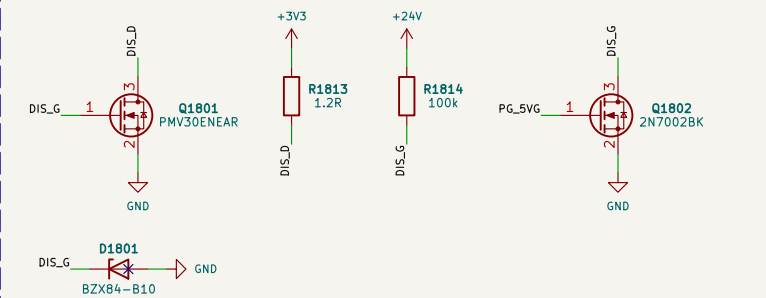
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52–4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5V0, FB 31.6k / 10k 0.1 % $\rightarrow$ 3.240–3.433 V ($\geq$ 3.20 V for the control board's reference), SS/TR 470p C0G; ramp-up 12–16 mV/us (F28003x VDDIO 8–100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control $\leq$ 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5V0 low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +V3 falls at ≥ 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 19 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic	
Sheet: /16 Live 5 V / 3.3 V, SELV connector/ File: 16_supplies.kicad_sch	
Title: PV-PWR power board – Live 5 V / 3.3 V, SELV connector	
Size: A3 KiCad E.D.A. 10.0.6	Date: 2026-10-05 Rev: A2 Id: 19/23

TAP A_IN+



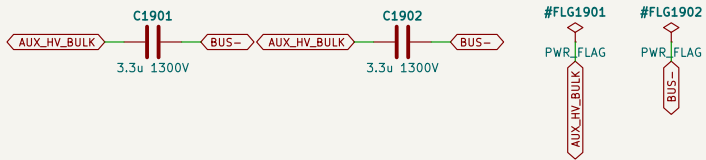
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

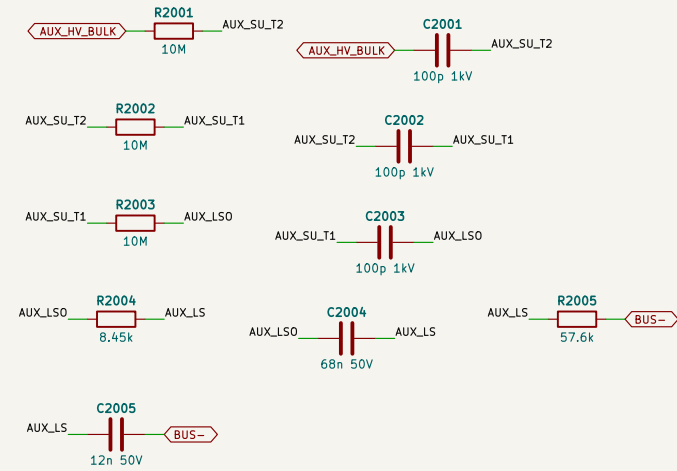
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 AUX75 input taps, OR, film/
File: 17_aux75_input.kicad_sch

Title: PV-PWR power board – AUX75 input taps, OR, film

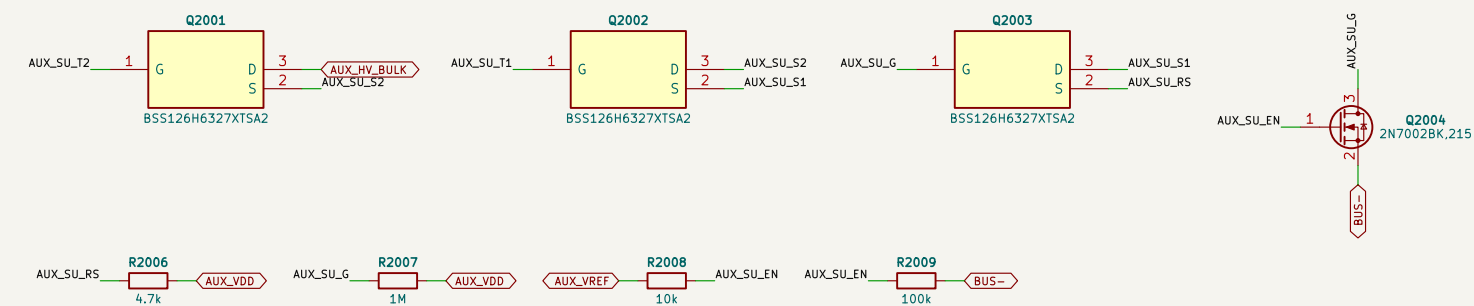
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 20/23

COMPENSATED TAP STRING



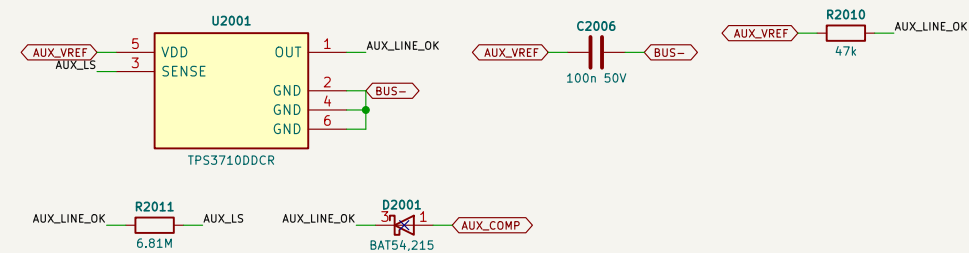
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



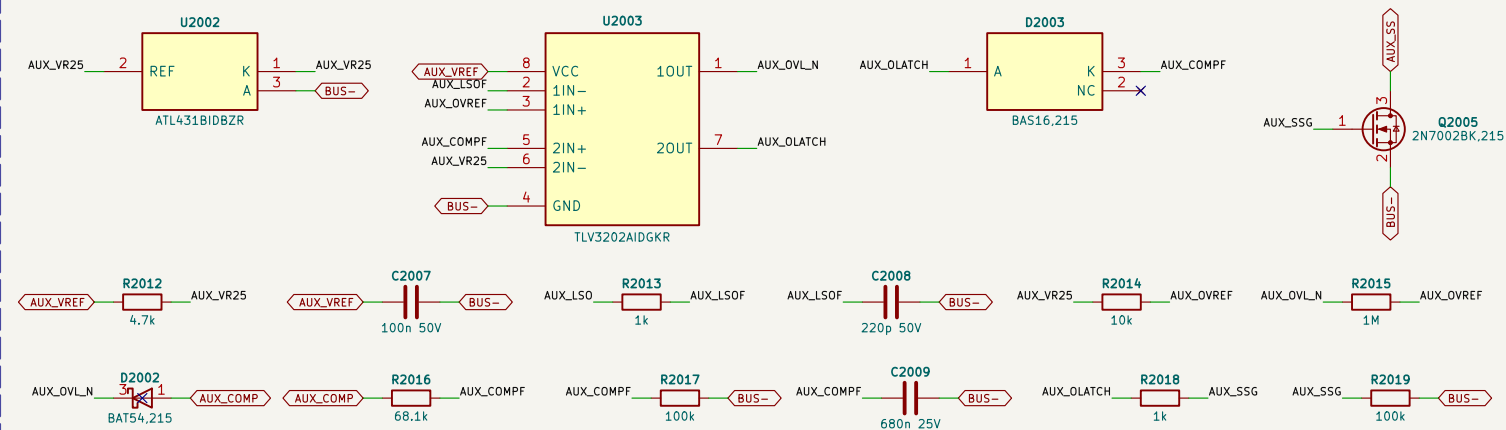
VREF high → start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

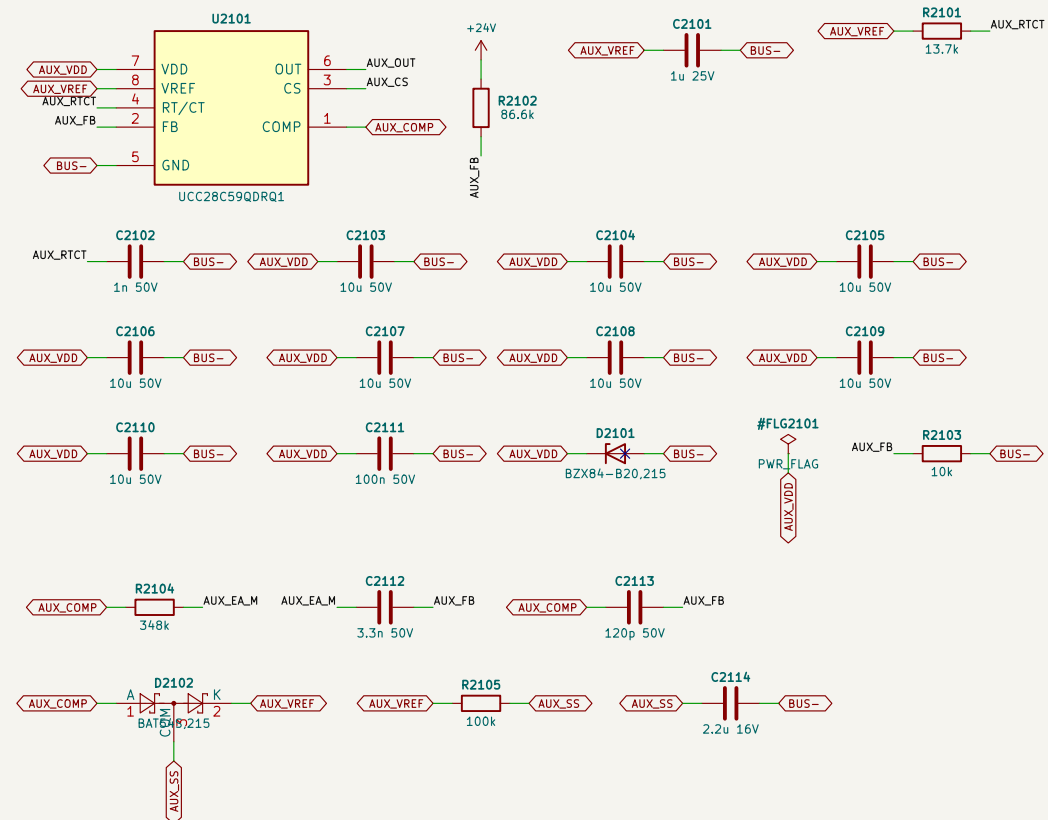
INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35-80 ms -> latch -> SS crowbar -> hiccup.

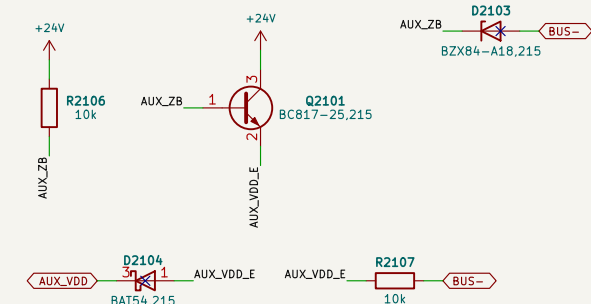
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 21 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic	
Sheet: /18 AUX75 start-up, brown-in, OV, timer/ File: 18_aux75_startup.kicad_sch	
Title: PV-PWR power board – AUX75 start-up, brown-in, OV, timer	
Size: A3	Rev: A2
Date: 2026-10-05	Id: 21/23
KiCad E.D.A. 10.0.6	

PWM CONTROLLER AND ERROR AMPLIFIER



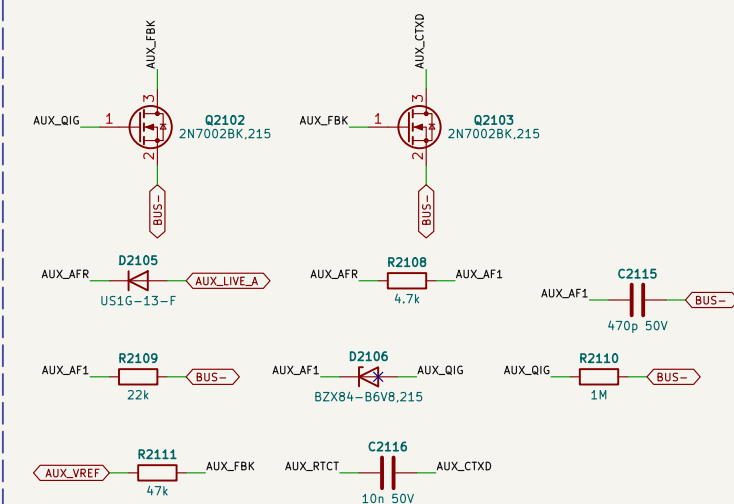
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



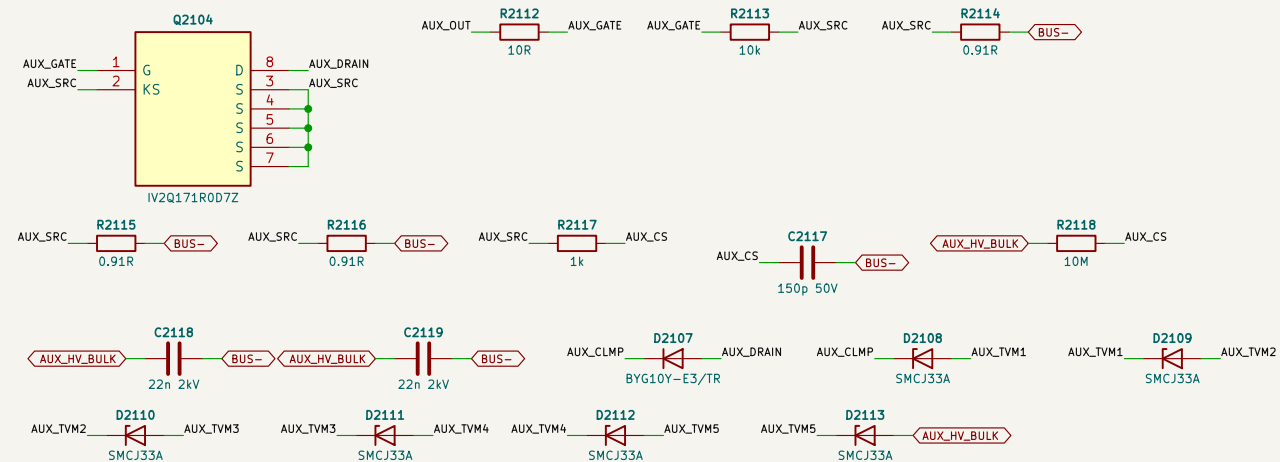
(30 W case, unchanged) follower Zener bias: $V_{DD} = \text{gate drive } 15.46\text{--}18.11 \text{ V}$ (BZX84-A18, $-40..85 \text{ }^{\circ}\text{C}$)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



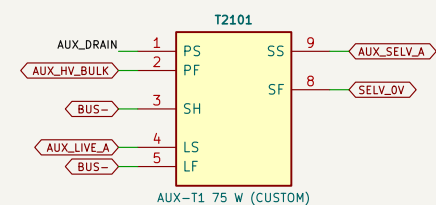
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



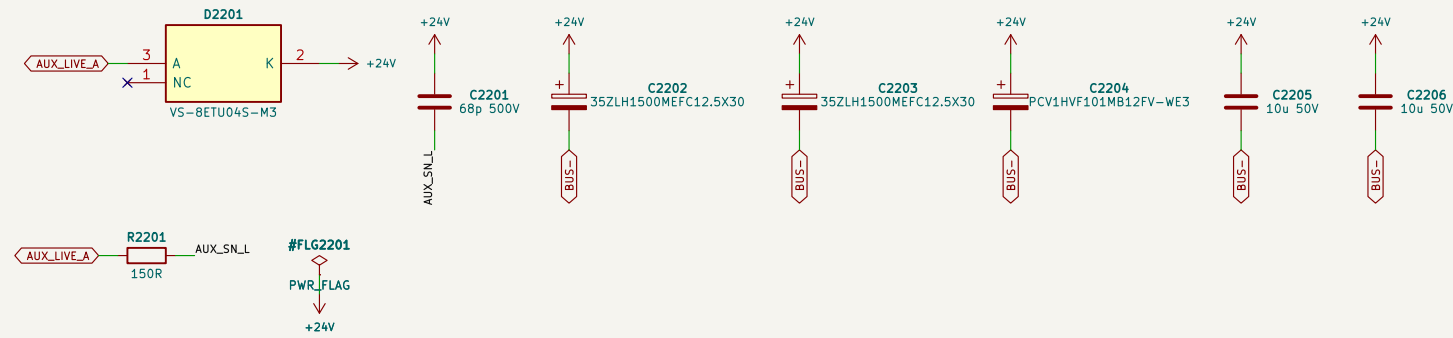
3 In parallel = 0.303333R; lowest limit 2.90 A delivers 103 W ($\geq 110\%$ of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK $\rightarrow$ CS (CRH2512); CS offset 101 mV at 1000 V cancels most of the 155–250 nA CS-jitter overshoot, so the highest current limit no longer rises with the input
Clamp: BYG10Y + 6 x SMCJ34; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



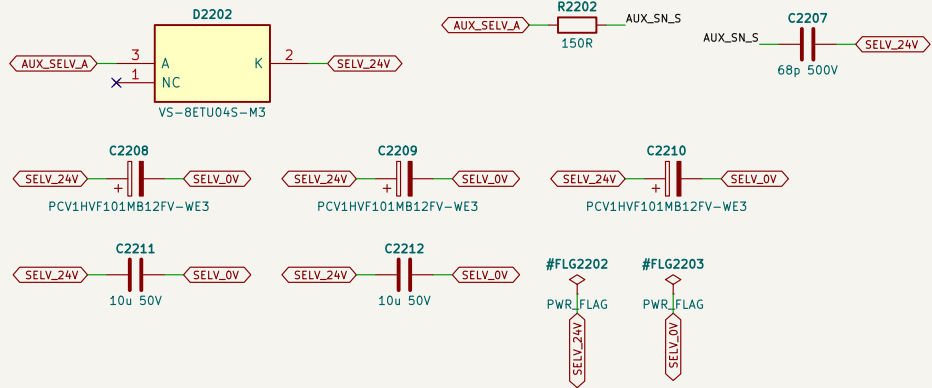
Lp 0.485 mH +/-7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

LIVE 24 V RECTIFIER, HOLD-UP



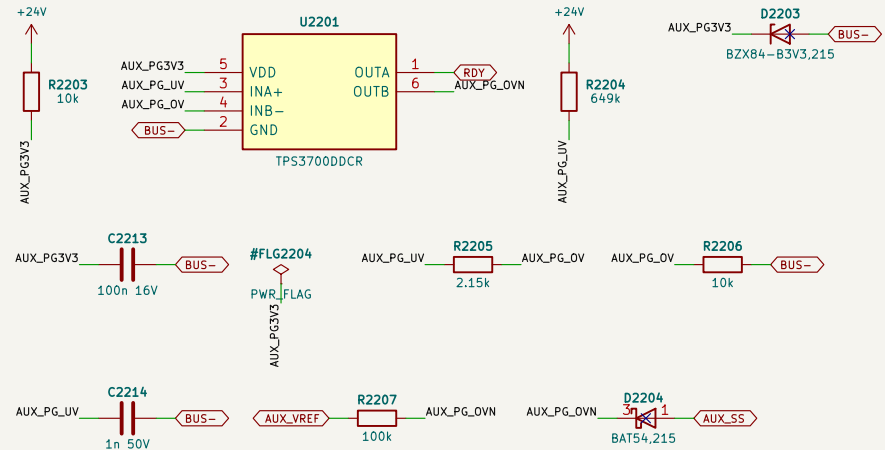
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)